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## Hormonal Modifiers of Mineral Metabolism Other Than Parathyroid Hormone, Vitamin D, and Calcitonin

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*Disorders of Mineral Metabolism, Vol. II*  
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ISBN 0-12-135302-8

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## I. ADRENAL GLUCOCORTICIDS

### A. Mode of Action of Steroid Hormones

Steroid hormones are ubiquitous, affecting all tissues either directly or indirectly. Even though different classes of steroid hormones may differ in their target cells and bring about different physiologic responses, their mode of action seems similar (Baxter, 1976).

Steroid hormones in the plasma generally are bound to specific transport proteins. For example, cortisol is bound to a specific  $\alpha$ -globulin termed the corticosteroid-binding globulin. Protein bound steroids probably serve as a reservoir while the circulating free hormones, which usually account for less than 10% of the total plasma concentration (Thompson and Lippman, 1974), represent the biologically active forms.

Steroid hormones can readily penetrate the cell membrane that appears devoid of specific receptors. Nevertheless, some hormones such as the glucocorticoids interact with plasma membranes and modify their properties. Inside the cell steroid hormones bind to specific cytoplasmic protein receptors. The formation of the steroid-cytosol receptor complex leads to a conformational change in the receptor. As a result, the complex migrates to the nucleus where it becomes bound to binding sites that are specific for the type and tissue of origin. This binding initiates transcriptional and translational events that result in the synthesis of proteins responsible for the steroid action (Gorski and Gannon, 1976).

The effects of these hormones may be either anabolic or catabolic, and target tissues are profoundly influenced by either an excess or lack of the effectors. For example, biochemical differentiation of several systems in fetal life may require glucocorticoids whereas the same steroids inhibit growth in the same tissue later in life. In addition to a role in biochemical differentiation and cell growth, induction of specific proteins in differentiated cells is a prominent physiologic and pharmacologic effect of steroid hormones (O'Malley, 1971). Another aspect of steroid function, particularly of the glucocorticoids, is their "permissive" action in allowing other nonsteroidal hormones to express their metabolic function fully (Thompson and Lippman, 1974).

## B. Interaction with Bone

Glucocorticoids have a pronounced effect on bone metabolism (Storey, 1963; Gallagher *et al.*, 1973a). Cushing's syndrome and the exogenous administration of glucocorticoids are both associated with decreased bone growth in children (Follis, 1951; Blodgett *et al.*, 1956; Iannaccone *et al.*, 1960) and growing animals (Yasumura *et al.*, 1976; Sheagren *et al.*, 1977), and marked loss of bone volume in the adult (Follis, 1951; Iannaccone *et al.*, 1960; Isaksson and Lindholm, 1965). Fracture healing is also impaired during cortisone therapy (Blunt *et al.*, 1950).

The glucocorticoids inhibit cartilage growth and development *in vitro* as demonstrated by the inhibition of sulfation of cartilage and decreased accumulation of acid mucopolysaccharides (Schryver, 1965; Barrett *et al.*, 1966). In addition, an inhibition of the incorporation of radioactive amino acids into cartilage collagen has been demonstrated (Daughaday and Mariz, 1962). Studies also suggest that

there is impaired synthesis of the organic matrix of bone and decreased new bone formation (Reynolds, 1966; Peck *et al.*, 1967; Oikarinen, 1977). Cortisol was found to decrease the incorporation of radioactive amino acids into collagen in bone culture (Peck *et al.*, 1967). Cortisol also decreased the incorporation of radioactive labeled nucleosides into RNA (Peck *et al.*, 1969). Evidence that the latter antianabolic effect of cortisol is dependent on protein synthesis is provided by the fact that the simultaneous administration of protein synthesis inhibitors abolished this action of cortisol (Peck *et al.*, 1969).

In addition to this antianabolic effect of glucocorticoids, a direct effect on bone cell development has been demonstrated. High-affinity glucocorticoid-binding proteins have been found in bone cells from fetal rat calvaria (Feldman *et al.*, 1975; Manolagas *et al.*, 1979). After steroid binding, the steroid-receptor complex is transferred from the cytosol and bound to the nucleus (Chen *et al.*, 1977). Glucocorticoid administration results in impaired bone cell growth, development, and maturation (Epker, 1970; Jee *et al.*, 1970; Jett *et al.*, 1970; Liskova-Kiar, 1979). Progesterone, an antagonist of glucocorticoids, partially blocked the glucocorticoid inhibition of bone cell growth (Chen *et al.*, 1977). There appears to be decreased mesenchymal cell differentiation into osteoclasts and osteoblasts. The result is a diminished number of bone cellular elements. This effect may be dose- and time-dependent. In short-term studies, high doses of cortisol (25–50 mg/kg/day) suppressed osteoprogenitor cell proliferation and bone formation. Lower doses (0.5–5.0 mg/kg/day) initially resulted in increased osteoprogenitor cell proliferation and increased numbers of osteoclasts (Jee *et al.*, 1970). With chronic glucocorticoid excess, regardless of dose, the number of bone cells was decreased (Jett *et al.*, 1970).

Measurements of skeletal dynamics in man, dog, and rabbit with Cushing's syndrome, either endogenous or exogenous, complement these findings. Bone formation is markedly suppressed as measured histologically (Jett *et al.*, 1970; Bressot *et al.*, 1976), autoradiographically (Rigg *et al.*, 1966), by tetracycline labeling (Jowsey and Riggs, 1970), or in radiocalcium kinetic studies (Bohr and Dawids, 1964). The suppression of bone formation is most apparent when bone is examined after prolonged glucocorticoid excess, thereby allowing for a new steady-state remodeling system of the basic metabolic unit of bone cells (osteon) to occur (Jett *et al.*, 1970). The degree of this suppressive effect on bone formation appears to be more severe with higher doses of glucocorticoids.

In addition to decreasing bone formation, the glucocorticoids also affect bone resorption (Storey, 1961; Riggs *et al.*, 1966; Jee *et al.*, 1970; Jett *et al.*, 1970; Bressot *et al.*, 1976). Cortisol in doses of from 0.5 to 5.0 mg/kg/day causes increased mesenchymal cell proliferation and an increased number of osteoclasts (Jee *et al.*, 1970). Bone resorption is stimulated as evidenced by accelerated removal of previously incorporated strontium 85 in the rat (Jee *et al.*, 1970), and the presence of increased resorption surfaces in bone from patients with en-

ogenous or exogenous Cushing's syndrome (Riggs *et al.*, 1966; Jett *et al.*, 1970; Bressot *et al.*, 1976). As this effect appears to be abolished by parathyroidectomy, the elevated levels of parathyroid hormone seen following glucocorticoid administration may mediate the increase in bone resorption (Jee *et al.*, 1970). A direct stimulatory effect is also possible since glucocorticoids have been reported to enhance bone resorption by peritoneal macrophages *in vitro* in the rat (Teitelbaum *et al.*, 1981). Glucocorticoids have also been observed to maintain the content of cytosol receptors for  $1,25(\text{OH})_2\text{D}_3$  in bone that might enhance bone resorption (Manolagas *et al.*, 1979). Jett *et al.*, (1970) concluded after studying the effects of cortisol on Haversian bone that, although an initial "burst" of osteoclasts may occur, both resorption and accretion are decreased. These investigators felt that the increase in resorption surfaces was due to an early rise in osteoclastic activity and to subsequent failure of new bone formation in areas where previous bone resorption had occurred. They also concluded that there was an uncoupling of bone resorption and formation and that, although both parameters were impaired, bone formation was slowed more than bone resorption, thus leading to a net increase in the rate of bone loss. At higher levels of cortisol administration (25–50 mg/kg/day), bone resorption is inhibited (Yasumura, 1976). Histologically, there initially occurs a decrease in osteoprogenitor cell proliferation (Jee *et al.*, 1970). *In vitro* studies also demonstrate that glucocorticoids inhibit bone resorption induced by parathyroid hormone (Stern, 1969; Liskova-Kiar, 1979).

### C. Interaction with Kidney

Glucocorticoids are required for some renal functions, such as free water clearance to proceed optimally (Kleeman *et al.*, 1975). Cortisol does not affect the acute renal handling of calcium or magnesium (Massry and Coburn, 1973). However, during prolonged glucocorticoid excess hypercalciuria and hypermagnesuria are seen whether or not plasma concentrations are normal or decreased (Clark *et al.*, 1959; Pechet *et al.*, 1959; Laake, 1960; Molinatti *et al.*, 1960; Collins *et al.*, 1962; Wajchenberg *et al.*, 1969). The chronic administration of cortisol or mineralocorticoids results in plasma volume expansion, increased renal blood flow and glomerular filtration rate, with a consequent increase in the renal excretion of sodium, calcium, and magnesium (Massry and Coburn, 1973; Kleeman *et al.*, 1975). Dietary sodium restriction in the experimental animal reverses this trend and leads to reduction in urinary calcium excretion (Suki *et al.*, 1968). In addition to chronic volume expansion, the soft tissue and skeletal breakdown seen in chronic glucocorticoid excess is also thought to contribute to the hypercalciuria (Kleeman *et al.*, 1975).

Glucocorticoids have been observed acutely to decrease urinary phosphate excretion (Persky *et al.*, 1955; Goldsmith *et al.*, 1965). During chronic

glucocorticoid administration, however, urinary phosphate excretion has been found to be high (Ingbar *et al.*, 1951; Roberts and Pitts, 1953; Anderson and Foster, 1959). Both the increase in urinary phosphate excretion and an increase in urinary cyclic AMP excretion in prednisone-treated patients can be suppressed with calcium infusion (Lukert and Adams, 1976). An increase in the plasma concentration of parathyroid hormone, as described below, may therefore be responsible for the increase in renal phosphate excretion. Plasma phosphate concentration in prednisone-treated patients is not significantly different than normal (Lukert and Adams, 1976) suggesting an increased flow of skeletal phosphate into the extracellular pool.

Glucocorticoid deficiency is associated with a decrease in urinary calcium excretion in man and animals (Lamberg and Torst, 1964; Kenny, 1966). This may be related to a reduction in the renal blood flow or glomerular filtration rate (Kleeman *et al.*, 1975).

#### D. Interaction with Vitamin D and Intestine

The antagonistic effect glucocorticoids have on the intestinal action of vitamin D was recognized in 1960 (Harrison and Harrison, 1960). Glucocorticoids have been shown to decrease the intestinal absorption of calcium *in vivo* (Fischer and Hastrup, 1954; Fox *et al.*, 1978; Hahn *et al.*, 1979a) and the active transport of calcium *in vitro* (Kimberg *et al.*, 1971). The mechanism by which this occurs remains unclear. Studies of vitamin D metabolism in the cortisol-treated rat have led to conflicting results. The original studies concerning the effect of glucocorticoids on vitamin D metabolism demonstrated normal 25-hydroxylation of vitamin D<sub>3</sub> in the liver and normal renal  $\alpha$ -hydroxylation of 25-hydroxyvitamin D<sub>3</sub> [25(OH)D<sub>3</sub>] in the rat (Kimberg *et al.*, 1971). In subsequent studies both a decrease in renal 1,25-dihydroxyvitamin D<sub>3</sub> [1,25(OH)<sub>2</sub>D<sub>3</sub>] production in the rat (Edelstein and Noff, 1977) and an increase in renal  $\alpha$ -hydroxylase activity in the chick (Spanos *et al.*, 1977) have been reported. Studies in man suggest that chronic glucocorticoid excess does not affect production or degradation of 1,25(OH)<sub>2</sub>D<sub>3</sub> (Seeman *et al.*, 1980). Conversion of 1,25(OH)<sub>2</sub>D<sub>3</sub> in the intestine of cortisol-treated rats into a more polar, biologically inactive metabolite than 1,25(OH)<sub>2</sub>D<sub>3</sub> has also been postulated as causing the alteration in intestinal calcium transport (Miravet *et al.*, 1975).

The possibility that glucocorticoids interfere with vitamin D action at its target organ has also been investigated. Intestinal localization and nuclear binding of the vitamin D metabolites occur in cortisone-treated rats (Favus *et al.*, 1973a). The levels of intestinal 1,25(OH)<sub>2</sub>D<sub>3</sub> were observed to be normal or reduced (Lukert *et al.*, 1973; Miravet *et al.*, 1975; Edelstein and Noff, 1977) following cortisol administration. One of the direct physiologic effects of vitamin D, the production of an intestinal calcium-binding protein, was found to be either nor-

mal (Kimberg *et al.*, 1971; Krawitt, 1972) or decreased (Edelstein and Noff, 1977) following cortisol administration in the rat. Cortisol decreased both intestinal calcium absorption and calcium-binding protein in the vitamin D-treated rachitic chick (Feher and Wasserman, 1978). It was initially observed that the administration of vitamin D or its metabolites,  $25(\text{OH})\text{D}_3$  (Favus *et al.*, 1973a) and  $1,25(\text{OH})_2\text{D}_3$  (Favus *et al.*, 1973b) to cortisol-treated rats did not return intestinal calcium absorption to normal. Similar observations were made in the rachitic chick (Feher and Wasserman, 1978). Miravet *et al.* (1975), however, found  $1,25(\text{OH})_2\text{D}_3$  returned calcium absorption to normal in cortisol-treated vitamin D-deficient rats. Glucocorticoid-mediated decrease in intestinal calcium absorption in the pig was also reversed by  $\alpha$ -hydroxyvitamin D (Fox *et al.*, 1978).

Studies in man demonstrated that  $25(\text{OH})\text{D}_3$  was effective in improving calcium absorption in iatrogenic Cushing's syndrome (Caniggia and Gennari, 1973; Hahn *et al.*, 1979a). Similarly, physiologic doses of  $1,25(\text{OH})_2\text{D}_3$  and  $\alpha$ -hydroxyvitamin  $\text{D}_3$  in prednisone-treated patients resulted in enhanced intestinal calcium absorption (Klein *et al.*, 1977, Lindgren *et al.*, 1978). Evidence that cortisol may cause an end-organ resistance to vitamin D in man is the finding that the plasma concentration of  $1,25(\text{OH})_2\text{D}_3$  is normal or elevated in corticosteroid-treated patients (Hahn *et al.*, 1979b; Seeman *et al.*, 1980) and rats (Lukert *et al.*, 1973). Plasma  $25(\text{OH})\text{D}_3$  has been found to either be normal (Aloia *et al.*, 1974, Hahn *et al.*, 1977) or low (Klein *et al.*, 1977; Seeman *et al.*, 1980).

Glucocorticoids alter the absorption of other substances such as iron and galactose (Kimberg *et al.*, 1971) and phosphate (Fox *et al.*, 1981). A generalized defect in the intestinal mucosal cell may then result in nonspecific malabsorption of a number of substances including calcium.

Little work has been carried out on the effect of glucocorticoids on the metabolic action of vitamin D on bone. Cortisol was found to decrease the resorptive capacity of  $25(\text{OH})\text{D}_3$  in tissue culture (Raisz *et al.*, 1972) and vitamin D-induced ultrastructural changes in bone cells have been observed to be unaffected by cortisol (Weisbrode and Capen, 1976). Recently glucocorticoids were observed to maintain the stability of cytosol  $1,25(\text{OH})_2\text{D}_3$  receptors in bone (Manolagas *et al.*, 1979).

## E. Interaction with Parathyroid Hormone

Early investigation into the effect of corticosteroids on parathyroid gland function suggested that corticosteroids decreased parathyroid hormone secretion (Eliel *et al.*, 1965). This inference found further support in studies that showed pharmacologic doses of glucocorticoids decreased parathyroid gland volume and mitotic rate in rats (Hansson, 1967). The excretion of parathyroid extract as determined by a rat bioassay system was unaltered by large doses of prednisone,

indicating that glucocorticoids do not increase the degradation of parathyroid hormone (Eliel *et al.*, 1965). Glucocorticoids do have an antagonistic effect on the peripheral action of parathyroid hormone (Stoerk *et al.*, 1963; Eliel *et al.*, 1965; Raisz, 1965; Wajchenberg *et al.*, 1965; Margargal *et al.*, 1969; Gozariu *et al.*, 1971; Raisz *et al.*, 1972; Williams *et al.*, 1974). Cortisol will worsen hypocalcemia and tetany in hypoparathyroidism and lower the serum calcium concentration in hypoparathyroid patients on vitamin D or parathyroid extract (Stoerk *et al.*, 1963; Eliel *et al.*, 1965; Canas *et al.*, 1967). This effect should lead to a compensatory increase in parathyroid hormone secretion in the parathyroid-intact subject (Williams *et al.*, 1974). Glucocorticoid administration has been shown to result in elevated plasma concentrations of immunoreactive parathyroid hormone in rat and man without a simultaneous change in serum calcium concentration (Williams *et al.*, 1974; Fucik *et al.*, 1975; Kukreja *et al.*, 1976; Lukert and Adams, 1976; Hahn *et al.*, 1979a). This has been observed during both acute and chronic glucocorticoid therapy (Fucik *et al.*, 1975). The reason for the increase in plasma parathyroid hormone concentration remains unclear. Although it may be secondary to the peripheral antagonism of parathyroid hormone or vitamin D, a direct stimulation of parathyroid hormone secretion by cortisol in parathyroid gland culture has also been observed (Au, 1976).

Parathyroid hormone is capable of eliciting its physiologic effect in the adrenalectomized or cortisol-excess animal (Stoerk *et al.*, 1963; Williams *et al.*, 1974). Greater amounts of parathyroid hormone are needed to exert the same hypercalcemic effect in cortisol excess animals as compared to normal controls (Stoerk *et al.*, 1963; Margargal *et al.*, 1969). Conversely, a greater rise in the serum calcium concentration in response to parathyroid hormone is seen in adrenalectomized than in intact rats (Williams *et al.*, 1974).

#### F. Interaction with Calcitonin

Determinations of serum calcitonin concentration in glucocorticoid excess or deficiency states have not yet been reported. Exogenous cortisol administration appears to increase the calcitonin content of the thyroid gland in the rat (de-Guisepe and Rindi, 1969; Yasumura *et al.*, 1976). Since cortisone decreases the protein content of the thyroid gland (Yatvin *et al.*, 1966), this increase in calcitonin may be due to decreased secretion rather than increased synthesis. Decreased calcitonin secretion could result from the lowered serum calcium induced by pharmacologic doses of glucocorticoids as discussed above.

The hypocalcemic effects of exogenous porcine calcitonin in the rat is decreased by the prior administration of pharmacologic doses (50–100 mg/kg/day) of cortisol (Thompson *et al.*, 1968; Eliel *et al.*, 1971). No change was noted in its hypophosphatemic effect (Thompson *et al.*, 1974). In contrast, adrenalect-



tomy does not alter the hypocalcemic action of calcitonin (Thompson *et al.*, 1968; Meyer and Brinkley 1972), although recovery from hypocalcemia is delayed (Lineberry and Waite, 1978). Studies in the rat indicate normal physiologic function of endogenous calcitonin in states of either cortisol excess or deficiency (Talmage *et al.*, 1970; Talmage and Kennedy, 1970). The apparent contradictory effect of cortisol on the hypocalcemic action of exogenous calcitonin cited above with endogenous calcitonin may be explained by the lower dose of cortisol given (5 mg/kg/day) in the study by Talmage *et al.* (1970).

Calcitonin has been demonstrated to prevent bone loss partially during corticosteroid therapy in the rabbit (Thompson and Urist, 1973; Eliel *et al.*, 1971; J. S. Thompson *et al.*, 1972). This suggests that calcitonin may serve to protect against the catabolic effect pharmacologic amounts of glucocorticoids have on the skeleton.

### G. Glucocorticoid Excess

Pharmacologic amounts of glucocorticoids are commonly used in a variety of clinical disorders and, as described above, can result in significant clinical complications.

The inhibition of growth in children with Cushing's syndrome and those receiving glucocorticoids for disorders such as severe allergic disease, rheumatoid arthritis, and the nephrotic syndrome has been well documented (Follis, 1951; Blodgett *et al.*, 1956; Iannaccone *et al.*, 1960; Strickland *et al.*, 1972). Cortisol will diminish the rate of longitudinal growth and skeletal maturation within a few weeks of onset of therapy (Blodgett *et al.*, 1956). While pharmacologic amounts of glucocorticoids are needed to suppress growth in endocrine intact children, physiologic amounts have a growth-suppressing effect in hypopituitary patients (Blodgett *et al.*, 1956); this may indicate an antagonism of glucocorticoids to the action of growth hormone. Studies in rats have demonstrated that the suppression of linear growth by corticosteroids or adrenocorticotrophic hormone (ACTH) can be overcome by administration of growth hormone (Becks *et al.*, 1944; Selye, 1952; Soyka and Crawford, 1965). Greater amounts of growth hormone were needed to offset the growth-suppressing effect of higher doses of glucocorticoids; this suggests a dose-related effect. Similarly, the optimum growth hormone effect in hypopituitary patients can be blunted with physiologic doses of cortisol (Soyka and Crawford, 1965). Prolonged administration of human growth hormone did not result in any improvement in growth rate, or retention of nitrogen, potassium, or phosphate in eight corticosteroid-treated children studied by Morris *et al.* (1968b). Others (Matiassevich and Gershberg, 1966) have reported some improvement in growth in similar patients while a positive nitrogen balance was obtained in a child with Cushing's syndrome after the administration of human growth hormone (Bergental and Lipsett, 1960).

Abnormal growth hormone secretion has also been proposed as a factor in the diminished growth rate in Cushing's syndrome. A diminished or absent response of growth hormone secretion to insulin-induced hypoglycemia has been reported in Cushing's syndrome secondary to a hypothalamic-pituitary disorder or adrenal tumors (Frantz and Rabkin, 1964; Hartog *et al.*, 1964; Krieger and Glick, 1972). Patients and normal volunteers receiving long-term corticosteroid administration for nonendocrine diseases also show similar abnormalities in growth hormone dynamics (Frantz and Rubkin, 1964; von Werder *et al.*, 1971; R. G. Thompson *et al.*, 1972). When glucocorticoid status was restored to normal in volunteers in patients with high dose exogenous glucocorticoids and in those whose Cushing's syndrome was secondary to adrenal tumors, growth hormone response to insulin-induced hypoglycemia returned to normal. Those patients with bilateral adrenocortical hyperplasia continued to lack normal growth hormone responsiveness following therapy (Krieger and Blick, 1972). This suggests a central abnormality is present in these patients. In contrast, studies of steroid-treated children have shown growth hormone dynamics to be normal (Root *et al.*, 1967; Morris *et al.*, 1968a; Strickland *et al.*, 1972).

In the adult, the disturbance in mineral metabolism most frequently manifests itself as osteoporosis. Cushing's syndrome is not infrequently associated with an increased incidence of fractures, particularly of the vertebrae (Demartini *et al.*, 1952; Curtiss *et al.*, 1954; McAllen, 1970). Findings of decreased total body calcium and phosphorus in Cushing's syndrome (Aloia *et al.*, 1974) complement the histologic picture described above and explain the propensity toward fractures encountered in these patients.

Patients with disorders such as rheumatoid arthritis, asthma, nephrosis, and systemic lupus erythematosus frequently are treated with pharmacologic amounts of glucocorticoids for prolonged periods of time (Demartini *et al.*, 1952; Soffer and Bader, 1952; Curtiss *et al.*, 1954; Klein *et al.*, 1965; McAllen, 1970). Although these patients appear to exhibit a greater than normal incidence of osteoporosis and fractures, the incidence of osteoporosis in patients treated with glucocorticoids does not seem to differ from that in patients who did not receive steroid therapy (McConkey *et al.*, 1962; Saville and Kharmosh, 1967). One study (Saville and Kharmosh, 1967) did report increased osteoporosis in female patients over the age of 50 who received glucocorticoids, but not in younger females or males. Thus it appears that age, sex, and underlying disease contribute to the development of osteoporosis. Although photon absorptiometry demonstrated bone loss during prednisone therapy over a 12-week period (Deding *et al.*, 1977), in another study in which five patients were treated with glucocorticoids *in vivo*, neutron activation analysis revealed no evidence of bone loss over a 20- to 33-month treatment period (Hosking and Chamberlain, 1973). It was suggested that when immobilization is not prominent in the disease, osteoporosis is a less common feature.

## H. Glucocorticoid Insufficiency

Glucocorticoid insufficiency is quite uncommon and if untreated will rapidly lead to the demise of the patient. Little information is therefore available concerning the effects of chronic glucocorticoid deficiency on mineral metabolism. In 1911 adrenalectomy was first shown to ameliorate tetany in parathyroidectomized animals (Guleke, 1911). Hypercalcemia following adrenalectomy has been demonstrated in rabbits (Kisch, 1924), cats (Taylor and Caven, 1927), dogs (Rogoff and Stewart, 1928; Myers and Lawrence, 1958), and rats (Raman, 1970a). Hypercalcemia has also been observed in patients with Cushing's syndrome and in Addison's disease after adrenalectomy (Loeb, 1932; Leeksma *et al.*, 1957; Walser *et al.*, 1963; Jorgenson, 1973).

The cause of the hypercalcemia in adrenal insufficiency is unclear. Actual elevation of serum ionized calcium has not been documented in man. Walser *et al.* (1963) were also unable to show an elevation in the plasma free calcium ion concentration in adrenalectomized dogs. Raman (1970a) did find elevated plasma ionized calcium levels in adrenalectomized rats, although others (Walser *et al.*, 1963) were unable to demonstrate any change in plasma ionized calcium concentration in this animal. Several mechanisms probably contribute to the development of hypercalcemia. Sodium and water loss lead to hemoconcentration with subsequent elevated plasma proteins and elevated total serum calcium concentration (Walser *et al.*, 1963; Raman, 1970a). When the ionic strength of plasma is decreased, there is a greater affinity of plasma proteins for calcium (Walser *et al.*, 1963). Hyponatremia may therefore result in an increase in the total plasma calcium concentration. Increased amounts of filterable and nonfilterable complexes of calcium with phosphate and citrate have also been found (Walser *et al.*, 1963). It seems unlikely that the parathyroid gland plays a role because of the occurrence of hypercalcemia in parathyroidectomized, adrenalectomized dogs (Myers and Lawrence, 1961). Increased intestinal absorption of calcium has not been shown to be a significant factor in the hypercalcemia of adrenal insufficiency (Rubin and Krick, 1936; Tibbets and Aub, 1937; Walser *et al.*, 1963). The possibility of an increased flux of calcium out of bone into the extracellular fluid remains (Myers and Lawrence, 1961; Schartum and Nichols, 1961; Canas *et al.*, 1967). *In vitro* studies have demonstrated that physiological concentrations of glucocorticoids decrease osteoclast activity without altering function and suggest that during adrenal insufficiency an increase in osteoclast activity may be responsible for the hypercalcemia seen in this disorder (Wong, 1979).

## I. Summary

The effect of glucocorticoid excess on mineral metabolism is manifest primarily by its influence on the skeleton. These steroid hormones inhibit both cartilage

and bone growth and development. In addition, loss of bone volume in the adult is seen. Bone formation is decreased by impaired organic matrix synthesis and also a decrease in mesenchymal cell differentiation into mature bone cells. Increased bone resorption and/or uncoupling of bone turnover with resorption exceeding formation may also be observed during glucocorticoid therapy.

The enhanced skeletal resorption appears to be secondary to an increase in parathyroid hormone secretion. During glucocorticoid therapy or in Cushing's syndrome a negative calcium balance is incurred. A decrease in intestinal calcium absorption may be secondary to alteration in vitamin D metabolism, impairment of the effect of vitamin D or its metabolites on the gut, or a nonspecific effect of glucocorticoid excess on intestinal absorption. In addition, hypercalciuria is seen and together with the increased fecal calcium excretion, is associated with an increased parathyroid hormone secretion.

Glucocorticoid insufficiency can result in hypercalcemia. This may be secondary to increased affinity of plasma protein for calcium, increased complexes of calcium with phosphate and citrate, plasma volume contraction, and an actual increase in ionized calcium.

## II. GONADAL STEROIDS

### A. Mode of Action

The mode of action of the gonadal steroids is similar to that of the glucocorticoids. They provide an example of a steroid hormone-mediated growth response. The first physiologic effects of these hormones are on the development of secondary sex tissues during embryogenesis. The effect of such hormones on secondary sex tissue in the adult is also well appreciated.

Although the skeleton, intestine, and kidney are not primary sites of action for the gonadal steroids, the interaction of sex hormones with these tissues has been under investigation because of the profound alterations in mineral metabolism in states of sex steroid deficiency. The mechanisms by which the gonadal steroids influence mineral metabolism is yet unclear. These effects will be discussed in the subsequent sections.

### B. Interaction with Bone

The effect of the sex steroids on bone metabolism has been under intense investigation ever since the hypothesis was put forth that senile osteoporosis was related to the menopause (Albright *et al.*, 1941). Following natural or artificial menopause, radiographic morphometry (Nordin *et al.*, 1966; Meema and Meema, 1968; Nordin, 1971; Horsman *et al.*, 1977; Recker *et al.*, 1977),

photodensitometry (Meema *et al.*, 1965; Aitken *et al.*, 1973a; Recker *et al.*, 1977), photon absorption (Davis *et al.*, 1966; Aitken *et al.*, 1972; Riggs *et al.*, 1981; Recker *et al.*, 1977; Lindsay *et al.*, 1978a), and total body neutron activation analysis (Aloia *et al.*, 1977; Chestnut *et al.*, 1977) techniques have demonstrated an accelerated loss of bone and/or a decrease in bone mass. Estrogen and/or androgen therapy in these patients results in a slowing or cessation of this loss of bone (Davis *et al.*, 1966; Aitken *et al.*, 1973b; Meema *et al.*, 1975; Chestnut *et al.*, 1977; Recker *et al.*, 1977; Lindsay *et al.*, 1978a, 1980; Christiansen *et al.*, 1980). Similar changes were also seen following progestogen therapy alone (Lindsay *et al.*, 1978a). An actual increase in bone mass was seen following estrogen therapy when the treatment was started within 3 years of the onset of the menopause (Aitken *et al.*, 1973b). It is during this period that the greatest loss of skeletal mass is seen (Davis *et al.*, 1966; Aitken *et al.*, 1973a; Horsman *et al.*, 1977). Withdrawal of estrogen therapy in ovariectomized (Lindsay *et al.*, 1978b) or postmenopausal (Horsman *et al.*, 1979) patients resulted in an accelerated loss of bone mineral comparable to that seen in patients who did not receive estrogen therapy. Christiansen *et al.* (1981), however, found that the rate of bone loss in postmenopausal women after discontinuation of estrogen/gestagen treatment was similar to an untreated control group.

Albright *et al.* (1941) originally noted that following the menopause there occurred a decrease in bone mass with a subsequent increase in fracture rate. He hypothesized that sex hormones were needed for optimal osteoblastic function and bone matrix synthesis. An estrogen-induced increase in the organic and mineral component of medullary bone in birds supports the concept that estrogens have an anabolic effect on bone (Turner and Schraer, 1977). Sex steroids also have been shown to have an effect on collagen metabolism. Estrogen has been demonstrated to inhibit collagen breakdown in rat uterus (Woessner, 1969), while both estrogen and testosterone were shown to have anabolic actions on bone collagen in the rat (Kowalewski *et al.*, 1971). A change in the lysine and hydroxylysine content of the insoluble bone collagen from rats treated with estrogens has been reported (Igarashi *et al.*, 1976). Studies demonstrating an improved calcium balance along with a decreased incidence of fractures and loss of height in postmenopausal women treated with estrogen convincingly demonstrated a role of estrogens in the maintenance of a normal skeleton (Wallach and Henneman, 1959; Hernberg, 1960; Gordan *et al.*, 1973; Lindsay *et al.*, 1980; Weiss *et al.*, 1980). However, the hypothesis of a decreased synthesis of bone matrix as the primary abnormality in postmenopausal osteoporosis is controversial. In 1958, radiocalcium kinetic studies demonstrated normal bone formation in patients with osteoporosis (Heaney and Whedon, 1958). This has been confirmed by others (Nordin, 1959; Lafferty *et al.*, 1964; Melick, 1966). Bronner *et al.* (1963), using similar techniques, however, demonstrated lower bone turnover in women with postmenopausal osteoporosis as compared to patients without

osteoporosis. Skeletal accretion rates utilizing stable strontium have shown either normal (Fraser *et al.*, 1960; Lafferty *et al.* 1964) or low values (Eisenberg and Gordan, 1961). Estrogen or androgen therapy in these patients resulted in a further lowering of bone accretion rates after prolonged treatment (Gordan and Eisenberg, 1963; Lafferty *et al.*, 1964). Tetracycline-labeling studies have usually shown low bone formation rates in postmenopausal and senile osteoporosis (Villanueva *et al.*, 1966; Wu and Frost, 1969). Increased bone resorption was demonstrated by the kinetic studies of Gordan and Eisenberg (1963), Lafferty *et al.* (1964), and Joly *et al.* (1980) in postmenopausal women with osteoporosis which was decreased with androgen or estrogen therapy. Subsequent calcium-kinetic studies have confirmed an increase in skeletal resorption in postmenopausal women as compared to perimenopausal or estrogen-treated postmenopausal women (Heaney *et al.*, 1978b). Quantitative morphologic analyses of bone have also demonstrated an increase in bone resorption in sex steroid deficiency states (Bordier *et al.*, 1965; Jowsey *et al.*, 1965; Wu *et al.*, 1967; Brown *et al.*, 1974; Bressot *et al.*, 1976; Joly *et al.*, 1980); however, no difference was noted in age-matched patients with and without osteoporosis (Meunier *et al.*, 1973). Estrogen has been shown to retard this process (Riggs *et al.*, 1969).

The mechanism by which the sex steroids slow bone loss during estrogen deficiency is uncertain. While estrogens appear to have no local effect on bone cells (Nutik and Cruess, 1974; Atkins and Peacock, 1975), concentrations of these steroids two- to tenfold greater than the concentrations that inhibit parathyroid hormone-stimulated calcium release, from bone, cause a net accretion of calcium *in vitro* (Stern, 1969; Atkins and Peacock, 1975). The most probable mechanism is the antagonistic effect of estrogens on the action of parathyroid hormone on bone as discussed below. The decrease in urinary excretion of hydroxyproline with sex steroid therapy would support this contention (Young *et al.*, 1968; Riggs *et al.*, 1969; Gallagher and Nordin, 1972; Lindsay *et al.*, 1976, 1978a). Estrogens also antagonize the peripheral effects of growth hormone as determined by the fall in urinary calcium and hydroxyproline when administered to patients with acromegaly (Schwartz *et al.*, 1969a) or to patients receiving growth hormone therapy (Schwartz *et al.*, 1969b).

There may be other, as yet unidentified, direct effects of estrogen on bone, as suggested by the observation that estrogens may induce tetany in idiopathic hypoparathyroidism (Burckhardt *et al.*, 1975). Hypercalcemia has also been reported to occur following estrogen withdrawal in vitamin D-treated hypoparathyroidism (Verbeelen and Fuss, 1979).

### C. Interaction with Kidney

Patients with postmenopausal or senile osteoporosis excrete normal amounts of calcium in the urine, but the fact that they are in negative calcium balance

(Henneman and Wallach, 1957) suggests that there may be an inappropriate urinary loss of calcium. As discussed below there is an improved calcium balance following estrogen or androgen therapy, brought about largely by a decrease in urinary calcium output.

The effect of testosterone and estrogen in decreasing urinary calcium excretion in postmenopausal osteoporosis was noted originally by Reifstein and Albright (1947). A number of subsequent studies have confirmed their observations (Shorr, 1945; Bogdonoff *et al.*, 1954; Henneman and Wallach, 1957; Young *et al.*, 1968; Gallagher *et al.*, 1973b; Heaney *et al.*, 1978a). Young and Nordin (1967) observed a rise in urinary calcium following natural or artificial menopause. They showed that estrogen administration caused a lowering of urinary calcium excretion to levels similar to those of the premenopausal state (Young *et al.*, 1968). A recent study demonstrated a significant inverse relationship of urinary calcium to endogenous estradiol production in oophorectomized women (Lindsay *et al.*, 1977).

The principal mechanism by which the sex steroids decrease urinary calcium excretion is thought to be secondary to decreased bone resorption. This results in a lowered serum calcium concentration and a reduced filtered load of calcium (Young *et al.*, 1968; Riggs *et al.*, 1969; Aitken *et al.*, 1971; Gallagher *et al.*, 1973b). This does not exclude a concomitant effect of these hormones on the kidney. Evidence for estrogen receptors in the kidney has been reported (Christy and Shaver, 1974). The administration of estrogens does not result in any significant change in renal blood flow or glomerular filtration rate (Christy and Shaver, 1974; Katz and Lindheimer, 1977), despite a significant decrease in urinary calcium excretion. It does appear that estrogens play a direct role in increasing sodium and water reabsorption (Christy and Shaver, 1974). Since there is a close relationship between sodium and calcium clearance (Massry *et al.*, 1973), estrogens may have a direct effect on renal tubular calcium transport via their action on sodium reabsorption in the proximal tubule.

Estrogen also affects phosphate metabolism. Elevated serum phosphate levels are frequently seen in estrogen-deficient states (Young and Nordin, 1967; Lindsay *et al.*, 1977) and can be lowered with estrogen therapy (Young *et al.*, 1968; Aitken *et al.*, 1971). Increased urinary phosphate excretion in patients with postmenopausal osteoporosis (Young and Nordin, 1967) is found to decrease with estrogen therapy (Bogdonoff *et al.*, 1954; Young *et al.*, 1968). In one of these studies no change in the phosphate excretion index was noted (Young *et al.*, 1968), whereas in another study urinary phosphate excretion increased following estrogen therapy and was accompanied by an elevated index of phosphate excretion (Aitken *et al.*, 1971). The divergent results can be explained by an influence of calcium on the renal handling of phosphate (Lavender and Pullman, 1963), since dietary calcium in these studies was either not monitored (Aitken *et al.*, 1971) or varied widely (Young *et al.*, 1968). The maximal renal tubular reabsorption of phosphate was found to be elevated in

estrogen deficiency (Lindsay *et al.*, 1977) and in another study in postmenopausal patients, stilbesterol lowered the renal tubular maximum for phosphate (Nassim *et al.*, 1956). Increased levels of urinary cyclic AMP have been found in oophorectomized women receiving mestronal (Lindsay *et al.*, 1976). These observations support the concept that changes in urinary phosphate excretion are secondary to changes in parathyroid hormone secretion.

#### D. Interaction with Intestine

Intestinal calcium absorption has usually been found to be low in patients with postmenopausal or senile osteoporosis (Bronner *et al.*, 1963; Caniggia *et al.*, 1963; Jaworski *et al.*, 1963; Munck, 1964; Spencer *et al.*, 1964; Avioli *et al.*, 1965; Parsons *et al.*, 1968; Gallagher *et al.*, 1973b, 1979) but not significantly different than in age- and sex-matched normal individuals (Avioli *et al.*, 1965; Bullamore *et al.*, 1970; Gallagher *et al.*, 1973b; Ireland and Fordtran, 1973). This abnormality may represent merely an effect of aging. Additional evidence that malabsorption of calcium may be present in many elderly individuals is the observation that a calcium intake as high as 1.5 gm may be required to produce a positive calcium balance (Heaney *et al.*, 1974, 1978a). This is approximately twice the usual dietary calcium intake in this population.

Although there has been no evidence to indicate that estrogens or androgens have a direct effect on intestinal calcium transport, these steroids do appear to affect calcium balance. Estrogens and androgens have been observed to improve calcium balance in patients with postmenopausal or senile osteoporosis (Reifenstein and Albright, 1947; Anderson, 1950; Bogdonoff *et al.*, 1954; Henneman and Wallach, 1957; Young *et al.*, 1968; Caniggia and Gennari, 1972; Gallagher *et al.*, 1973a; Heaney *et al.*, 1978b). It has been demonstrated in sheep that estrogens increase the efficiency of absorption of dietary calcium (Braithwaite *et al.*, 1972). Similarly, the intestinal absorption of radioactive calcium and stable calcium has been shown to improve following the administration of sex steroids to patients with postmenopausal osteoporosis (Jaworski *et al.*, 1963; Caniggia and Gennari, 1972; Gallagher *et al.*, 1980). In other studies, however, little effect on these parameters were noted despite achievement of a positive calcium balance (Gallagher *et al.*, 1973b; Spencer *et al.*, 1976; Recker *et al.*, 1977). Finally, twice-elevated values for intestinal calcium absorption have been found in pregnancy (Heaney and Skillman, 1971). Although estrogens may be responsible, in part, for this observation, placental lactogen has also been implicated (Heaney and Skillman, 1971). As discussed in a later section, prolactin is also capable of increasing intestinal calcium absorption.

Impaired intestinal absorption of phosphate has also been reported in postmenopausal women (Caniggia and Gennari, 1972). Estrogen therapy was observed to improve intestinal phosphate absorption (Young *et al.*, 1968; Caniggia



and Gennari, 1972), and phosphate balance (Bogdonoff *et al.*, 1954). If, as discussed below, estrogens can increase the synthesis of  $1,25(\text{OH})_2\text{D}_3$ , the expected end result would be enhanced absorption of both calcium and phosphate.

### E. Interaction with Parathyroid Hormone

Parathyroid hormone has been shown to be required for the development of immobilization osteoporosis (Burkhart and Jowsey, 1967) and calcium-deficient osteoporosis (Jowsey and Raisz, 1968). Because of the striking preponderance of osteoporosis in the postmenopausal or estrogen-deficient female (Meema *et al.*, 1965; Davis *et al.*, 1966; Nordin *et al.*, 1966), the interaction of sex steroids with parathyroid hormone has received considerable attention. The concentration of serum immunoreactive parathyroid hormone in estrogen-deficient patients has usually been found to be normal or low (Gallagher *et al.*, 1973b; Riggs *et al.*, 1973; Brown *et al.*, 1974; Teitelbaum *et al.*, 1976; Aloia *et al.*, 1977; Davies *et al.*, 1977; Joly *et al.*, 1980) although a number of patients with postmenopausal osteoporosis have been found to have elevated levels of serum parathyroid hormone (Riggs *et al.*, 1973; Berlyne *et al.*, 1974; Teitelbaum *et al.*, 1976; Joly *et al.*, 1980). These divergent results may reflect assay specificity, variable renal function, early primary hyperparathyroidism, or a subpopulation of these patients. The finding of increased resorption lacunae in postmenopausal and hypogonadal patients with osteoporosis (Bordier *et al.*, 1965; Jowsey, 1966; Brown *et al.*, 1974; Bressot *et al.*, 1976; Joly *et al.*, 1980) does suggest increased parathyroid hormone activity.

There is experimental evidence that the sex steroids may modulate the peripheral effects of parathyroid hormone. *In vivo* studies have shown an increased sensitivity of bone to parathyroid hormone in ovariectomized rats (Orimo *et al.*, 1972). In two *in vitro* bone culture systems, estrogens were found to decrease the bone resorptive response to parathyroid hormone (Stern, 1969; Atkins *et al.*, 1972). However, in a third study no effect of  $17\beta$ -estradiol on parathyroid hormone resorption was demonstrated (Caputo *et al.*, 1976). Estrogens have been shown to interfere with the metabolic effects of parathyroid hormone on bone cells as manifested by both reduced glycolysis and accumulation of citric acid (Atkins *et al.*, 1972); this may explain the antagonism to the skeletal effect of parathyroid hormone. Liskova (1976) observed that ethinyl estradiol decreased calcium release from parathyroid hormone-treated bone as well as control bone and concluded the inhibitory action was caused by a toxic effect on bone cells. Progesterone, testosterone, and androstenedione have also been observed to inhibit the resorptive action of parathyroid hormone in bone culture (Atkins and Peacock, 1975; Peacock *et al.*, 1977).

Further evidence that estrogens might antagonize the peripheral effects of parathyroid hormone is derived from clinical studies. Greater bone loss has been

found in postmenopausal women with primary hyperparathyroidism as compared to premenopausal primary hyperparathyroid patients (Hossain *et al.*, 1970). Ethinyl estradiol was found to decrease bone resorption and urinary hydroxyproline excretion in postmenopausal women with primary hyperparathyroidism (Gallagher and Wilkinson, 1973). Estrogen therapy in ovariectomized patients resulted in a fall in the serum calcium and a rise in the serum parathyroid hormone concentration (Riggs *et al.*, 1972a; Gallagher *et al.*, 1980). This effect of estrogens on the serum calcium concentration has been applied successfully in the treatment of postmenopausal women with primary hyperparathyroidism (Gallagher and Wilkinson, 1973) and patients with parathyroid carcinoma (Goepfert *et al.*, 1966; Sigurdsson *et al.*, 1973b). Testosterone also has been found to be effective in lowering the serum calcium concentration in primary hyperparathyroidism due to parathyroid carcinoma or adenoma (Landau and Kappas, 1965).

#### F. Interaction with Calcitonin

The observation that the rise in the serum calcium concentration following an oral calcium load was greater in female than in male rats suggests that either there is a sex difference in calcitonin secretion or in its peripheral action (Sorenson and Hindberg, 1971; Peng and Garner, 1980). Thyroidectomy has been shown to result in a rise in the serum calcium concentration in the female rat, which was prevented by prior oophorectomy (Foster *et al.*, 1977). These results suggest that estrogens antagonize the effect of calcitonin. Estrogens have been shown to antagonize the hypocalcemic effects of exogenously administered calcitonin in the rabbit (Currie and Black, 1972) and rat (Shai and Wallach, 1972). Androgens have been observed to enhance the hypocalcemic response to calcitonin in the rat (Hinde and Phillipppo, 1968; Ogata *et al.*, 1970; Sorenson and Hindberg, 1971); however, in one study this was only true in castrated rats (Sorenson and Hindberg, 1971). In addition, a lack of endogenous androgen decreased calcitonin's hypocalcemic action (Ogata *et al.*, 1970; Hollo *et al.*, 1971). The effects of both androgens and estrogens on the hypocalcemic effects of calcitonin are thought to reflect an effect on bone.

Comparison of basal serum calcitonin concentrations in men and women has yielded conflicting results. Delorme *et al.* (1976) reported higher values in women than in men whereas others found opposite results (Heath and Sizemore, 1977; Hillyard *et al.*, 1978). Other studies found no sex differences in basal serum calcitonin concentration (Parthemore and Deftos, 1978; Deftos *et al.*, 1980). In the study of DeLorme *et al.* (1976), estrogen-deficient women had serum calcitonin levels that were similar to males, and estrogen therapy in these women returned the serum calcitonin concentration to the normal range for estrogen-sufficient women. In addition, there was a significantly greater rise in

the serum calcitonin concentration to calcium stimulus in the estrogen-replete women than in estrogen-deficient women. The ability of estrogen to increase the basal and calcium-stimulated serum calcitonin concentration has been confirmed by others (Hillyard *et al.*, 1978; Morimoto *et al.*, 1980; Stevenson *et al.*, 1980). Heath and Sizemore (1977) and Deftos *et al.* (1980) noted a greater rise in plasma calcitonin in response to both calcium and/or pentagastrin infusion in men as compared to premenopausal women, while Parthemore and Deftos (1978) found this sex difference only with pentagastrin. The reason for the differences between these studies is unclear.

### G. Interaction with Vitamin D

Patients with postmenopausal osteoporosis have decreased intestinal calcium absorption and frequently a negative calcium balance. Since vitamin D plays an important role in intestinal calcium transport, an interaction of sex steroids with vitamin D has been sought. Low serum concentrations of  $1,25(\text{OH})_2\text{D}_3$  in osteoporotic patients and elderly subjects (Gallagher *et al.*, 1979) along with a decreased conversion of  $25(\text{OH})\text{D}_3$  to  $1,25(\text{OH})_2\text{D}_3$  in aging rats (Armbrecht *et al.*, 1980) suggest impaired metabolism of vitamin D may contribute to the decrease in intestinal calcium absorption. One study however reported appropriate formation of  $25(\text{OH})\text{D}_3$  and  $1,25(\text{OH})_2\text{D}_3$  from isotopically labeled cholecalciferol in osteoporotic women (Davies *et al.*, 1977). Vitamin  $\text{D}_3$  has been found to increase calcium absorption in postmenopausal women, but only with doses over 20,000 units per day (Hall *et al.*, 1969; Gallagher *et al.*, 1973b). The effects of physiologic doses of  $1,25(\text{OH})_2\text{D}_3$  have been found to be normal in patients with postmenopausal osteoporosis (Davies *et al.*, 1977; Gallagher *et al.*, 1979). The concentration of plasma  $25(\text{OH})\text{D}_3$  in estrogen-deficient postmenopausal patients with osteoporosis is not different from that in age-matched normal individuals (Lund *et al.*, 1975; Gallagher *et al.*, 1979; Davies *et al.*, 1977). In one study the level of  $1,25(\text{OH})_2\text{D}_3$  was found to be significantly below normal in such patients (Gallagher *et al.*, 1976); however, in a similar study this finding was not confirmed (Haussler and McCain, 1978). Estrogen administration to postmenopausal women has been found to cause a rise in the plasma concentration of  $1,25(\text{OH})_2\text{D}_3$  with a simultaneous increase in intestinal calcium absorption (Gallagher *et al.*, 1980). The increased  $1,25(\text{OH})_2\text{D}_3$  levels could be secondary to the simultaneous increase in serum parathyroid hormone concentration (Gallagher *et al.*, 1980), or to a direct effect of estrogen itself on vitamin D metabolism. Kenny (1976) observed an increased synthesis of  $1,25(\text{OH})_2\text{D}_3$  in the Japanese quail during ovulation and later demonstrated that the *in vivo* administration of estrogen enhanced the *in vitro* synthesis of  $1,25(\text{OH})_2\text{D}_3$  by kidney homogenates (Baksi and Kenny, 1977). This finding confirmed similar observations by Tanaka *et al.* (1976) in the Japanese quail and in the chick.

Estrogen in these studies was found to increase  $1,25(\text{OH})_2\text{D}_3$  formation while decreasing the synthesis of 24,25-dihydroxyvitamin  $\text{D}_3$  [ $24,25(\text{OH})_2\text{D}_3$ ]. Estradiol has also been observed to stimulate *in vivo* production of  $1,25(\text{OH})_2\text{D}_3$  in the rat (Baksi and Kenny, 1978). Recent preliminary studies have shown that estradiol added to a chick kidney cell culture (Henry, 1978) decreases  $1,25(\text{OH})_2\text{D}_3$  production. This suggests that the *in vivo* effects of estrogens on  $1,25(\text{OH})_2\text{D}_3$  production are indirect. Progesterone has also been found to increase  $1,25(\text{OH})_2\text{D}_3$  production in the Japanese quail, but was less effective than estrogen (Baksi and Kenny, 1977). Testosterone itself does not cause a similar increase in  $1,25(\text{OH})_2\text{D}_3$  formation, but may be required to show the estrogen-enhanced increase of  $1,25(\text{OH})_2\text{D}_3$  synthesis in chicks (Tanaka *et al.*, 1976, 1978; Pike *et al.*, 1978). In the immature quail testosterone was not required to demonstrate an estrogen effect (Baksi and Kenny, 1977).

Testosterone has been found to augment the increased plasma levels of  $25(\text{OH})\text{D}_3$  in male and female rats during ultraviolet irradiation (Ohata *et al.*, 1977). Estrogens had no effect and neither testosterone nor estrogen increased  $25(\text{OH})\text{D}_3$  levels without irradiation.

## H. Summary

The influence of gonadal steroids on mineral metabolism is most obvious during sex steroid deficiency states. A characteristic example is in the postmenopausal woman where one observes an accelerated decrease in bone mass along with a slight elevation of the serum calcium concentration and increased urinary calcium and hydroxyproline excretion. Although decreased skeletal matrix formation may contribute to the decrease in skeletal mass, enhanced mineral resorption appears to be a major factor in the subsequent loss of bone. Since estrogen antagonizes the effects of parathyroid hormone on bone, the relative estrogen deficiency in the postmenopausal women probably leads to an increased skeletal sensitivity to parathyroid hormone and increased skeletal resorption. Estrogen administration can also increase intestinal calcium absorption in the postmenopausal woman. This effect may be mediated through an increased formation of  $1,25(\text{OH})_2\text{D}_3$ . The improvement in calcium absorption would decrease parathyroid hormone secretion and therefore decrease skeletal resorption.

## III. THYROID HORMONE

### A. Mode of Action

The molecular basis for thyroid hormone action is in considerable dispute. A large body of evidence has been gathered in support of the concept that hormonal

action initiates at the nuclear level (Oppenheimer *et al.*, 1976). This is based upon demonstration of specific high affinity binding sites in nuclei for triiodothyronine, the correlation of the number of nuclear sites in various tissues with the known effects of thyroid hormone on these tissues, the correlation between the relative biological potency of thyroid hormone analogs and the affinity of nuclear binding, and the limited number of nuclear sites that could modulate hormone response. As yet there is no convincing evidence that a specific cytosol-binding protein transports the hormone to the nucleus in a manner analogous to those of steroid hormones.

Other mechanisms of thyroid hormone action that have been proposed include activation of mitochondria, enhanced sensitivity of adrenergic receptors, and stimulation of Na,K-ATPase. It is conceivable that one or more of these mechanisms might be a secondary phenomenon to that of nuclear transcription.

## B. Interaction with Bone

The effects of thyroid hormone on bone have been extensively documented since the first description of bone disease in a patient with hyperthyroidism by von Recklinghausen (1891). In this initial report and subsequent studies histologic examination of bone in hyperthyroidism has consistently revealed evidence of increased bone resorption mediated by osteoclasts, particularly in cortical bone (Hunter, 1930; Askanazy and Rutishauser, 1933; Follis, 1953; Adams *et al.*, 1967; Bordier *et al.*, 1967; Meunier *et al.*, 1972; Bressot *et al.*, 1976; Mosekilde *et al.*, 1977b). An increase in fibrovascular bone marrow is also a feature of hyperthyroid bone disease. The presence of increased osteoclastic bone resorption and fibrous bone marrow are characteristic of the lesions of osteitis fibrosa in patients with hyperparathyroidism. However, there are morphological differences which allow separation of the two disorders. In hyperthyroidism, osteoclastic activity on cortical bone surfaces is greater than on trabecular bone surfaces (Meunier *et al.*, 1972; Mosekilde *et al.*, 1977a), whereas in hyperparathyroidism trabecular osteoclastic activity is much greater. Another feature that aids in distinguishing the two osteopathies is the size of the periosteocytic lacunae, an index of osteocytic osteolysis. In primary hyperparathyroidism the lacunae are increased in size, whereas in hyperthyroidism they are normal (Meunier *et al.*, 1972; Mosekilde *et al.*, 1977b).

Numerous foci of osteoblasts are often found near sites of osteoclastic activity. Because Follis (1953) noted osteoid seams in six patients, he felt that osteomalacia was a complication of thyrotoxicosis. Quantitative morphometry of bone from thyrotoxic patients has now clearly shown that this is not so (Bordier *et al.*, 1967; Bressot *et al.*, 1976; Mosekilde *et al.*, 1977b). Although there is an increased percentage of bone surface covered by osteoid, the osteoid thickness is not increased and the total volume of osteoid is normal or only slightly increased.

Studies of tetracycline labeling of bone indicate that the calcification rate of bone is increased (Bressot *et al.*, 1976; Mosekilde *et al.*, 1977b).

Even though bone resorption and bone formation have been shown by histologic means to have increased in hyperthyroidism, the degree of bone resorption appears to predominate in most patients. This is reflected by a decrease in both cortical and trabecular bone (Meunier *et al.*, 1972; Bressot *et al.*, 1976; Mosekilde *et al.*, 1977b).

Surgical treatment of hyperthyroidism (Meunier *et al.*, 1972) and antithyroid medication (Mosekilde and Melsen, 1978) have both led to an increase in bone mass.

The histologic abnormalities in hyperthyroidism are reflected by the radiologic features of the disorder. Most commonly there is evidence of decreased bone mass as manifested by increased radiolucency of the vertebral bodies, patchy demineralization of the cranial bones and a "flaky" appearance of the cortices in the phalanges, pelvis, and long bones (Meunier *et al.*, 1972). In general, the radiologic manifestations of osteoporosis in thyrotoxicosis are difficult to distinguish from those of osteoporosis of other causes. The occurrence of pseudofractures is extremely rare in hyperthyroidism (Meunier *et al.*, 1972); only two instances have been reported. Another uncommon finding specific to the hyperthyroidism of Grave's disease is the presence of periosteal overgrowth of the hands and feet (Gimlette, 1960; Kinsella and Back, 1968). This has been termed thyroid achropachy. Clubbing of the fingers and toes may be prominent in some patients. The pathogenesis of this interesting manifestation of Grave's disease is unknown, but is always associated with exophthalmos and usually with pretibial myxedema. It does not appear to arise from the increased secretion of thyroid hormones, since it typically occurs after treatment of the disease or after spontaneous remission.

Biochemical evidence of increased bone turnover in hyperthyroidism was first reported by Krane and colleagues (1956). After injection of radiocalcium they found that the specific activity of the labeled calcium in serum and urine declined more rapidly in patients with hyperthyroidism than in normal subjects. In addition there was an increased exchangeable pool of calcium and rates of calcium resorption from and deposition in bone were increased. These observations were confirmed in human (Bordier *et al.*, 1967) and animal studies (Aubert *et al.*, 1964).

Further evidence of increased bone resorption is provided by studies of urinary and serum hydroxyproline levels (Dull and Henneman, 1963; Kivirikko *et al.*, 1965; Mosekilde *et al.*, 1978). This amino acid, which is almost unique to collagen, has been an excellent marker of bone matrix resorption in a variety of bone diseases. Excretion of this amino acid is often increased in hyperthyroidism and, since serum levels have been found to be increased, a direct effect of thyroid hormone on renal excretion of hydroxyproline cannot account for the hydroxy-

prolinuria (Kivirikko *et al.*, 1965). Because increased turnover of skin collagen could also contribute to the amount of hydroxyproline excreted, Askenasi and Demeester-Mirkin (1975) measured a more specific marker of bone collagen resorption, the glucosyl-galactosyl-hydroxylysine/galactosyl-hydroxylysine urinary ratio. In normocalcemic hyperthyroid subjects the ratio was 1.5, virtually identical to normal. In two patients who were hypercalcemic, the ratio was less than one. This would indicate that a significantly greater proportion of the hydroxylysine had its origin in bone.

Elevation of serum alkaline phosphatase concentration is common in hyperthyroidism (Cook *et al.*, 1959; Smith *et al.*, 1973; Moseleilde *et al.*, 1978) and correlates reasonably well with urinary hydroxyproline excretion (Smith *et al.*, 1973). The enzyme elevation in most patients probably arises from increased osteoblastic activity. This is supported by the finding that the skeletal isoenzyme of alkaline phosphatase in serum is often increased in patients with hyperthyroidism (Richter and Ohlen, 1971; Cleeve and Brown, 1978). In severe thyrotoxicosis of long duration liver disease may arise and also lead to an elevated serum alkaline phosphatase.

The skeletal manifestations of hypothyroidism depend to a great extent on the age of onset of the disorder. In children, hypothyroidism characteristically leads to an arrest of growth, manifested radiologically by increased density of the provisional zone of calcification and a delay in the appearance of ossification of epiphyseal centers (Krane and Goldring, 1978). Less commonly it leads to a generalized osteopetrosis or marble bone disease (Braid, 1951; Fanconi and Girardet, 1952; Jeune and Béraud, 1955; Breton *et al.*, 1958; Jeune and Muller, 1959; Andersen, 1961; Royer and Mathieu, 1962; Schmidt, 1962; Tümay *et al.*, 1962). In some of the osteopetrotic patients, hypercalcemia, nephrocalcinosis, or soft tissue calcification was present (Braid, 1951; Fanconi and Girardet, 1952; Schmidt, 1962; Tümay *et al.*, 1962). In adult onset hypothyroidism, radiologic changes are not seen, but quantitative morphometry of bone has revealed a markedly decreased area of cancellous bone undergoing resorption (Bordier *et al.*, 1967). This was associated with an increase in cancellous bone volume. Radiocalcium kinetic studies in hypothyroidism have demonstrated a decreased calcium pool and low rates of bone resorption and formation (Krane *et al.*, 1956; Bordier *et al.*, 1967). Decreased collagen turnover is reflected by low levels of urinary hydroxyproline excretion (Kivirikko *et al.*, 1965). Serum alkaline phosphatase concentration may be low in children, probably as a result of decreased osteoblastic activity (Talbot, 1939).

The effects of excessive secretion of thyroid hormones on bone appear to be direct, since *in vitro* studies indicate that thyroxine and triiodothyronine can stimulate osteoclastic bone resorption (Mundy *et al.*, 1976) and increased bone resorption can be produced by thyroid hormone administration to parathyroidectomized dogs (Adams and Jowsey, 1967). As discussed in the section on thyroid

hormone interaction with parathyroid hormone, the modulation of intracellular cyclic AMP concentration may be an important effect of thyroid hormone.

### C. Interaction with Kidney

It has not been demonstrated that thyroid hormone has direct effects on renal handling of divalent cations and phosphate, but calcium and phosphate excretion are altered in abnormal thyroid states. Hypercalciuria is a frequent feature of hyperthyroidism but rarely leads to renal calculi formation or nephrocalcinosis (Bradley *et al.*, 1974). It is likely that the hypercalciuria results from increased mobilization of bone mineral (Krane *et al.*, 1956) and suppression of parathyroid hormone secretion (Buckle *et al.*, 1969; Bouillon and DeMoor, 1974; Burman *et al.*, 1976; Rude *et al.*, 1976; Mosekilde and Christensen, 1977). Both of these factors probably account for the hyperphosphatemia often found in hyperthyroidism (Fujita *et al.*, 1958; Cook *et al.*, 1959; Parsons and Anderson, 1964). Parathyroid hormone suppression leads to hyperphosphatemia as a result of increased renal tubular reabsorption of phosphate, a finding well documented in patients with hyperthyroidism (Harden *et al.*, 1964; Parsons and Anderson, 1964; Bijvoet and Majoor, 1965; Malamos *et al.*, 1969). In hypothyroidism there is a reduction of glomerular filtration (Bradley *et al.*, 1974) and generally less calcium is excreted in the urine than normal. However, there are documented cases of hypercalciuria and nephrocalcinosis in children (Naylor, 1955; Royer *et al.*, 1958; Jeune and Muller, 1959; Lang, 1960; Royer and Mathieu, 1962; Schmidt, 1962; Tümay *et al.*, 1962; Job *et al.*, 1963). This appears to occur as a consequence of intestinal hyperabsorption of calcium. Hypothyroidism is associated with a reduced renal tubular reabsorption of phosphate (Nordin and Fraser, 1956; Adams *et al.*, 1968) and a decreased tubular maximum for phosphate (Malamos *et al.*, 1969). Despite this, patients often have a normal or only slightly low serum phosphate concentration (Aub *et al.*, 1929; Adams *et al.*, 1968; Malamos *et al.*, 1969). It is possible that a reduction in glomerular filtration rate and intestinal hyperabsorption of phosphate counterbalance the renal phosphate leak.

### D. Interaction with Intestine

Studies in both animals and man indicate a striking influence of thyroid hormone on intestinal absorption of calcium. Although an excess of thyroid hormone is generally associated with increased metabolic activity, in the hyperthyroid state intestinal calcium absorption is usually reduced (Cook *et al.*, 1959; Finkelstein and Schacter, 1962; Friedland *et al.*, 1965; Adams *et al.*, 1967; Krawitt, 1967; Noble and Matty, 1967; Williams *et al.*, 1967; Bordier *et al.*, 1967; Shafer and Gregory, 1972; Singhelakis *et al.*, 1974; Swaminathan and



Care, 1975; Haldimann *et al.*, 1980). The converse has been found in most studies of hypothyroidism (Jeune and Muller, 1959; Lang, 1960; Lowe *et al.*, 1962; Royer and Mathieu, 1962; Lekkerkerker *et al.*, 1971; Lekkerkerker and Doorenbos, 1973), although in two rat studies increased calcium absorption was not found (Winter *et al.*, 1970; Hehrmann *et al.*, 1973). The increased intestinal absorption of calcium observed in hypothyroid children may contribute to the hypercalcemia found in some patients (Braid, 1951; Johnson and White, 1952; Naylor, 1955; Royer *et al.*, 1958; Jeune and Muller, 1959; Schmidt, 1962; Job *et al.*, 1963; Pretorius and Potgieter, 1971). In adults with hypothyroidism, hypercalcemia has been reported in only three patients (Lowe *et al.*, 1962; Adams *et al.*, 1968). There may be two reasons for the effects of thyroid hormone on intestinal calcium absorption. The hyperthyroid state is associated with a reduced intestinal transit time (Middleton, 1971). This could account for some calcium malabsorption. In turn the prolonged intestinal transit time noted in hypothyroidism could explain the more efficient absorption of calcium in this condition (Middleton, 1971).

A second possibility is that thyroid hormone might play a role in vitamin D metabolism. This is discussed in Section II,G.

#### E. Interaction with Parathyroid Hormone

There is little evidence that thyroid hormone acts directly on parathyroid hormone secretion. In patients with mild or frank hypercalcemia, the concentration of parathyroid hormone in the circulation is low (Buckle *et al.*, 1969; Bouillon and DeMoor, 1974; Burman *et al.*, 1976; Rude *et al.*, 1976; Mosekilde and Christensen, 1977). In three studies of normocalcemic hyperthyroid patients (Bouillon and DeMoor, 1974; Manderlier *et al.*, 1974; Castro *et al.*, 1975) parathyroid hormone concentration was normal but in one study (Bommer *et al.*, 1976) the levels were elevated in 15 out of 20 patients. Adult hypothyroid patients have been found to have high normal or elevated concentrations of parathyroid hormone, usually in the presence of a normal calcium concentration (Adams *et al.*, 1968; Bouillon and DeMoor, 1974; Castro *et al.*, 1975; Bouillon *et al.*, 1980). Further evidence of suppressed parathyroid hormone levels in hyperthyroid patients is provided by the finding of increased renal tubular reabsorption of phosphate in many patients (Bortz *et al.*, 1961; Harden *et al.*, 1964; Parsons and Anderson, 1964; Bijvoet and Majoer, 1965; Malamos *et al.*, 1969). Hyperphosphatemia is another possible manifestation of the secondary hypoparathyroidism of hyperthyroidism (Fujita *et al.*, 1958; Cook *et al.*, 1959; Parsons and Anderson, 1964). Bommer and colleagues (1976) feel that hyperphosphatemia is not a consequence of suppressed parathyroid function, since they observed hyperphosphatemia in the absence of hypercalcemia or low concentrations of parathyroid hormone. In hypothyroidism, hypophosphatemia is

seldom found but tubular reabsorption of phosphate is decreased (Adams *et al.*, 1968); this presumably reflects an increased concentration of parathyroid hormone in the circulation.

The peripheral metabolism of parathyroid hormone could conceivably influence hormone levels in the circulation of patients with hyperthyroidism or hypothyroidism by altering the rate of parathyroid hormone clearance. No studies have been reported that bear on this aspect of the interaction of parathyroid with thyroid hormone.

The coexistence of primary hyperparathyroidism and hyperthyroidism may lead to confusion in the differential diagnosis of hypercalcemia (Parfitt and Dent, 1970; Richards, 1970; Bergdahl, 1977) unless plasma levels of parathyroid hormone are shown to be elevated in hypercalcemic patients with both disorders (Burman *et al.*, 1976). The reversal of hypercalcemia by appropriate therapy of the hyperthyroidism provides definitive evidence of thyrotoxic hypercalcemia. Persistence of the hypercalcemic state after the patient is euthyroid strongly suggests the presence of primary hyperparathyroidism or another hypercalcemic disorder.

The sensitivity of patients with thyroid disease to exogenous parathyroid hormone has been examined in three studies (Harrison *et al.*, 1964; Castro *et al.*, 1975; Guttler *et al.*, 1977). In each study patients with hyperthyroidism were found to be more responsive to the hormone than normal subjects or hypothyroid patients. Serum calcium concentrations rose to higher levels in two studies (Harrison *et al.*, 1964; Castro *et al.*, 1975). Excretion of hydroxyproline and phosphate in the urine also were greater in hyperthyroid patients (Castro *et al.*, 1975). Urinary cyclic AMP excretion was greater in response to parathyroid extract in hyperthyroid patients than in normal subjects and was less in hypothyroid patients than in the normal group (Guttler *et al.*, 1977). The greater effect of parathyroid hormone on serum calcium concentration and urinary hydroxyproline in hyperthyroidism could be explained by a high prevailing rate of bone resorption (see section on bone) and/or by relatively low levels of endogenous parathyroid hormone in the circulation. The latter factor could also explain the greater phosphaturic and cyclic AMP responses. In the hypothyroid patients a decreased rate of bone resorption and/or a relatively larger concentration of circulating parathyroid hormone could explain the insensitivity to exogenous parathyroid hormone. It is possible that peripheral responsiveness to parathyroid hormone in states of abnormal thyroid function is an important factor in determining the secretory rate of the parathyroid glands. The study of urinary cyclic AMP excretion in thyroid hormone excess or deficiency states (Guttler *et al.*, 1977) suggests that the hormone modulates end-organ activity by an unknown interaction with membrane-bound adenylate cyclase. The *in vitro* observation that thyroid hormone can inhibit bone cyclic nucleotide phosphodiesterase (Mar-

cus, 1975) provides an alternate or complementary explanation for thyroid hormone action. Since propranolol can reverse the hypercalcemia of thyrotoxicosis (Rude *et al.*, 1976) thyroid hormone may act on bone through  $\beta$ -adrenergic modulation of intracellular cyclic AMP concentration.

#### F. Interaction with Calcitonin

A direct effect of thyroid hormone on calcitonin secretion has not been demonstrated, although antithyroid drugs have been reported to decrease the rate of secretion of calcitonin in pigs (Duncan and Care, 1967). In contrast to this study, Peng and colleagues have found that in rats iodine deficiency (Peng *et al.*, 1976) or propylthiouracil (Peng *et al.*, 1977) induces goiter formation, C-cell hyperplasia, and hypersecretion of calcitonin. Hypercalcemia has been noted in iodine deficient rats and may be primarily responsible for the stimulation of calcitonin (Clark *et al.*, 1978). The measurement of serum calcitonin concentrations in hyperthyroidism by sensitive techniques has not been reported but in a few patients with Grave's disease, the concentration of calcitonin in the thyroid gland was low (Aliopoulos *et al.*, 1966; Mazzuoli *et al.*, 1966).

The hypocalcemic effect of calcitonin is enhanced in patients with hyperthyroidism (Bijvoet *et al.*, 1968; Martin and Melick, 1969) and in rats treated with thyroid hormone (Milhaud *et al.*, 1967; Cherny *et al.*, 1970). A similar enhancement of calcitonin-induced hypocalcemia has been found in hypothyroid (thyroidectomized) rats (Talmage and Kennedy, 1969) and athyreotic children (Milhaud, 1968; Lambertz *et al.*, 1969). The sensitivity to calcitonin in the absence of the thyroid gland probably can be attributed to calcitonin deficiency.

#### G. Interaction with Vitamin D

In three studies of patients with thyrotoxicosis, plasma 25(OH)D levels were significantly lower than normal (Hunstein *et al.*, 1977; Mosekilde *et al.*, 1977a; Valenzas *et al.*, 1977). In one study the concentration was below the lower limit of normal in 11 of 12 patients who had elevated plasma alkaline phosphatase concentrations (Hunstein *et al.*, 1977). In three subsequent studies normal 25(OH)D levels were reported (Cleeve and Brown, 1978; Kaptein *et al.*, 1979; Bouillon *et al.*, 1980). The significance of these findings and of the lowered levels of this hepatic metabolite of vitamin D in certain patients remain unknown. In one study in patients with hypothyroidism, normal levels were reported. In two studies decreased levels of serum 1,25(OH)<sub>2</sub>D were reported (Kaptein *et al.*, 1979; Bouillon *et al.*, 1980). It is not known whether these results reflect a direct action of thyroid hormone on production rate or metabolic clearance rate of the vitamin D metabolite, effects of blood phosphate and parathyroid hormone on production

rate, or a combination of these influences. Elevated levels of  $1,25(\text{OH})_2\text{D}$  were found in one study of patients with hypothyroidism (Bouillon *et al.*, 1980). The precise mechanism which accounts for this has also not been determined.

Hypersensitivity to vitamin D as manifested by hypercalcemia has been observed in several hypothyroid children (Royer and Mathieu, 1962; Job *et al.*, 1963; Wilkins, 1965; Pretorius and Potgieter, 1971) as well as in hypothyroid dogs (Handovsky and Goormaghtigh, 1937). Resistance to the effect of vitamin D on reducing fecal calcium excretion was found in two adult hyperthyroid patients (Cook *et al.*, 1959). However, several other studies suggest that sensitivity to vitamin D may be present in some patients with hyperthyroidism. In a rare case of hypoparathyroidism and hyperthyroidism, vitamin D requirements for the control of hypocalcemia were reduced when the patient developed hyperthyroidism (Rothenberg and Redleaf, 1959). Further indirect evidence of a role of vitamin D in patients with hyperthyroidism is provided by the observation that corticosteroid therapy can suppress the hypercalcemia of thyrotoxicosis in some patients (Sataline *et al.*, 1962; Harper and Hughes, 1970; Richards, 1970).

## H. Summary

Thyroid hormone has been demonstrated to have profound effects on mineral metabolism. A direct effect on bone accounts for increased bone turnover in patients with hyperthyroidism and reduced bone turnover in hypothyroid patients. Hypercalcemia may occur in hyperthyroidism when the rate of bone resorption exceeds that of bone formation and is accompanied by reduced levels of serum parathyroid hormone, hyperphosphatemia, an elevated renal threshold for phosphate excretion, increased urinary hydroxyproline and hypercalciuria. An impairment in intestinal calcium absorption has also been noted, probably resulting from decreased intestinal transport time and reduced  $1,25(\text{OH})_2\text{D}$  levels. Hypothyroidism in children produces marked impairment of growth at the epiphyses. Rarely, hypercalcemia and/or hypercalciuria may occur in hypothyroid patients. This may occur as a result of increased intestinal calcium absorption and a reduced exchangeable calcium pool.

## IV. GROWTH HORMONE

### A. Mode of Action

Growth hormone is the major hormonal regulator of growth. Its effects include anabolism of cartilage and bone, lipolysis, and a decrease in glucose tolerance. The growth-promoting influence of growth hormone appear to be effected through the intermediary action of somatomedin as discussed in Section IV,B.

The mechanism by which growth hormone acts on fat and carbohydrate metabolism is less clear. A relationship with the adenylate cyclase-cyclic AMP system has been suggested (Goodman and Schwartz, 1974).

## B. Interaction with Bone

Growth hormone is a necessary factor for normal growth and development. Although growth hormone plays no role in fetal growth, its absence in childhood results in a growth rate only one-third to one-half that of normal (Daughaday, 1977). Excess growth hormone will result in accelerated linear growth in childhood leading to gigantism and to bony overgrowth in the adult (Daughaday, 1974).

Growth hormone has no direct effect either on cartilage (Salmon and Daughaday, 1957) or on bone (Canalis *et al.*, 1977b) *in vitro*. All anabolic effects on the skeleton appear to be mediated through the intermediary action of the somatomedins. The somatomedins are a group of peptides that circulate bound to another growth hormone-dependent carrier protein (Daughaday, 1977). These peptides appear to be synthesized and released by the liver since serum somatomedin activity falls following partial hepatectomy in the rat (Uthne and Uthne, 1972) and is low in cirrhosis (Wu *et al.*, 1974). In addition, perfusion of rat liver with growth hormone increases somatomedin activity (McCanoghey and Sledge, 1970; Phillips *et al.*, 1976). The action of somatomedin is to enhance the incorporation of proline into collagen, uridine into RNA, and thymidine into DNA (Salmon and duVall, 1970; Raisz *et al.*, 1976). There is good correlation between the rise in serum somatomedin activity and the growth response in hypopituitary children with growth hormone (Daughaday and Parker, 1963). The mechanism by which somatomedin acts on its target cells is unclear. Unlike many peptide hormones, it does not appear to act by stimulating an increase in cyclic AMP. Somatomedin has been found to decrease the basal level and parathyroid hormone-induced increase in chondrocyte adenylate cyclase (Tell *et al.*, 1973).

Although growth hormone is probably the major regulator of somatomedin production, other factors may influence its metabolism. Low levels of somatomedin are seen in fasting (Daughaday and Kipnis, 1966), protein-calorie malnutrition (Grant *et al.*, 1973; Beaton, 1976), and renal failure (Saenger *et al.*, 1974; Lewy and New, 1975). Insulin (Daughaday *et al.*, 1976) and prolactin (Francis and Hill, 1975) have been reported to increase somatomedin production. Modification of the end-organ response to the effect of somatomedin may be present on an inherited basis as in the Laron dwarf (Daughaday *et al.*, 1969) or during starvation (Salmon, 1975).

Growth hormone increases the skeletal uptake of radioactive calcium in the rat (Ulrich *et al.*, 1952) and dog (Harris and Heaney, 1969), and of stable strontium

in man (Eisenberg and Gordan, 1961). Similarly, in acromegaly the skeletal uptake of radioactive calcium is increased (Haymovitz and Horwith, 1964; Aloia *et al.*, 1972; Montz *et al.*, 1973). When measured simultaneously, both bone accretion and resorption are increased in patients with acromegaly (Bell and Bartter 1967; Aloia *et al.*, 1972) and during growth hormone therapy (Haas *et al.*, 1976), as determined by radioactive calcium kinetic studies. Microradiographic studies have also indicated that bone formation and resorption are increased in acromegaly (Roelfsema *et al.*, 1970; Riggs *et al.*, 1972b), although one study demonstrated normal bone formation (Aloia *et al.*, 1972). High bone turnover in acromegaly or with growth hormone therapy is also suggested by the finding of increased urinary excretion of hydroxyproline (Jasin *et al.*, 1962; Smiley and Ziff, 1964; Wright *et al.*, 1965; Aloia *et al.*, 1976; Haas *et al.*, 1976).

The growth hormone-induced changes in bone formation and resorption may not be closely coupled throughout cortical and trabecular bone. Quantitative morphometry has generally shown a net increase in periosteal new bone formation, while in some studies resorption appears to predominate in intracortical or Haversian bone (Frost, 1966; Kruse and Kuhlencordt, 1975). Endosteal bone resorption is increased in man (Frost, 1966; Riggs *et al.*, 1972b), but the dog has been reported to have a net increase in endosteal bone formation in response to growth hormone (Harris *et al.*, 1972).

Osteoporosis was thought to be a manifestation of acromegaly (Daughaday, 1974), based on radiographic findings of skeletal rarefaction (Scriver and Bryan, 1935). The lack of an increased incidence of compression fractures of the spine in acromegalic patients and a number of morphometric and kinetic studies of bone in acromegaly do not support this concept. Eisenberg and Gordan (1961) demonstrated an increase in the miscible calcium pool in acromegaly through the use of stable strontium. These findings were confirmed by radiocalcium kinetic studies in which exchangeable calcium was found to be either increased (Haymovitz and Horwith, 1964; Aloia *et al.*, 1972; Montz *et al.*, 1973) or normal (Bell and Bartter, 1967). In addition, bone mineral content has been found to be normal (Doyle, 1967; Montz *et al.*, 1973), while bone density (Roelfsema *et al.*, 1970; Haas *et al.*, 1976) as well as bone mass (Riggs *et al.*, 1972b) have been reported to be normal or increased. Quantitative microradiographic studies demonstrated cortical and trabecular bone thickness to be normal or increased (Roelfsema *et al.*, 1970; Aloia *et al.*, 1972). Similar studies by Riggs *et al.* (1972b) also demonstrated an increase in cortical bone thickness, but found trabecular bone thickness to be decreased.

### C. Interaction with Kidney

Acromegaly is associated with hypercalciuria, which decreases to normal with treatment of the pituitary tumor (Scriver and Bryan, 1935; Bauer and Aub, 1941;

Harrison *et al.*, 1960; Molinatto *et al.*, 1961; Nadarajah *et al.*, 1968; Sjöberg, 1969; Sigurdsson *et al.*, 1973a). In addition, growth hormone administration has been observed to result in an increase in urinary calcium excretion (Ikkos *et al.*, 1958; Hutchings *et al.*, 1959; MRC, 1959; Beck *et al.*, 1960; Henneman *et al.*, 1960; Hanna *et al.*, 1961; Karam *et al.*, 1961; Forbes *et al.*, 1962; Wright *et al.*, 1965; Melvin *et al.*, 1967; Bryson *et al.*, 1972). The exact cause of the hypercalciuria is unclear. The administration of growth hormone to dogs was observed to have no effect on renal calcium or phosphate clearance, renal blood flow, or glomerular filtration rate over a 2-hr period following its administration (Westby *et al.*, 1977). This suggests that acute changes in the plasma growth hormone concentration do not influence calcium or phosphate homeostasis. In contrast, the glomerular filtration rate and renal plasma flow have been shown to be high in acromegaly (Heller *et al.*, 1954; Ikkos *et al.*, 1956; Gershberg *et al.*, 1957) and following chronic exogenous growth hormone administration in man (Corvilain *et al.*, 1962) and dog (White *et al.*, 1949; Davis *et al.*, 1954; Corvilain *et al.*, 1964). As discussed below, growth hormone therapy may be associated with enhanced intestinal calcium absorption. This could contribute to the increased urinary calcium excretion. A recent study has shown a significant positive correlation between intestinal calcium absorption and renal calcium excretion (Sigurdsson *et al.*, 1973a). It is also clear that the hypercalciuria that occurs under the influence of growth hormone may be in excess of that which can be accounted for by increased intestinal calcium absorption (Scriver and Bryan, 1935; Bauer and Aub, 1941; Ikkos *et al.*, 1958; Beck *et al.*, 1960; Nadarajah *et al.*, 1968; Bryson *et al.*, 1972). This suggests another effect of growth hormone on the kidney and/or bone. Fraser *et al.* (1960) found that parathyroidectomy prevented the hypercalciuric effect of growth hormone in rats, a finding which suggests that the hypercalciuria might be related to increased parathyroid hormone secretion. More recently, however, an increase in urinary calcium excretion following growth hormone administration was seen in hypoparathyroid patients (Hernandez *et al.*, 1967; Bryson *et al.*, 1972). An effect of growth hormone in decreasing renal tubular citrate concentration has been suggested as playing a role in the hypercalciuria (Karam *et al.*, 1961). As discussed above, increased bone turnover with increased skeletal calcium resorption also may be a contributing factor in the hypercalciuria.

Although early studies suggested growth hormone caused increased urinary phosphate excretion (Scriver and Bryan, 1935; Bauer and Aub, 1941), subsequent investigations have generally indicated a reduced urinary phosphate excretion (MRC, 1959; Gershberg, 1960; Henneman *et al.*, 1960; Corvilain *et al.*, 1962; Forbes *et al.*, 1962; Bryson *et al.*, 1972). The phosphate excretion index has either been normal or decreased (Camanni *et al.*, 1968; Nadarajah *et al.*, 1968). The mechanism for decreased renal phosphate excretion appears to include a direct effect of growth hormone on the kidneys, since parathyroidectomy does not alter this effect in man (Bryson *et al.*, 1972) or dog (Corvilain *et*

*al.*, 1964). As indicated earlier, the increased tubular reabsorption of phosphate is associated with an increased serum phosphate concentration and positive phosphate balance.

#### D. Interaction with Intestine

Increased intestinal absorption of calcium and/or low fecal calcium excretion have been found in some patients with acromegaly (Bauer and Aub, 1941; Hanna *et al.*, 1961; Sigurdsson *et al.*, 1973a) or following exogenous growth hormone administration (Beck *et al.*, 1957; Ikkos *et al.*, 1958; Henneman *et al.*, 1960; Hanna *et al.*, 1961; Bryson *et al.*, 1972; Sigurdsson *et al.*, 1973a). In sheep, growth hormone also appears to enhance intestinal calcium absorption (Braithwaite, 1975). However, in some patients growth hormone produces no change or an increase in fecal calcium excretion (Scriver and Bryan, 1935; Ikkos *et al.*, 1958; Forbes *et al.*, 1962; Melvin *et al.*, 1967; Bryson *et al.*, 1972). A number of studies have also shown negative calcium balance in acromegaly or during growth hormone therapy (Scriver and Bryan, 1935; Bauer and Aub, 1941; Ikkos *et al.*, 1959; Beck *et al.*, 1960; Hanna *et al.*, 1961; Bell and Bartter, 1967; Nadarajah *et al.*, 1968), although many reports of a positive calcium balance have also been published (Scriver and Bryan, 1935; Shorr *et al.*, 1953; Beck *et al.*, 1957; Hutchings *et al.*, 1959; Henneman *et al.*, 1960; Pearson *et al.*, 1960; Nadarajah *et al.*, 1968; Sjöberg *et al.*, 1969; Montz *et al.*, 1973). These conflicting results may be related to the amount of dietary calcium ingested, since increasing dietary calcium intake from low to high levels may convert a negative calcium balance to a positive balance (Scriver and Bryan, 1935; Pearson *et al.*, 1960). Other possible contributing factors toward a negative calcium balance include increased intestinal calcium secretion (Heaney and Skillman, 1964) and hypercalciuria.

Growth hormone deficiency produced by hypophysectomy resulted in decreased intestinal calcium absorption in rats, which returned toward normal with growth hormone therapy (Finkelstein and Schachter, 1962; Mainoya, 1975b). When calcium transport was corrected for intestinal weight, however, no change in intestinal calcium absorption was seen in hypophysectomized rats (Krawitt *et al.*, 1977). Under similar experimental conditions, while hypophysectomy did not prevent the increase in active calcium transport seen in response to a low calcium diet, there was a significant decrease in intestinal calcium absorption when compared to normal rats (Kimberg *et al.*, 1961). In these studies, a comparable defect in glucose transport was not seen; this suggests that the defect in calcium absorption is not secondary to a nonspecific effect of hypophysectomy on the intestine. As discussed below, hypophysectomy in the rat results in decreased levels of  $1,25(\text{OH})_2\text{D}_3$  in the blood and could account for decreased intestinal calcium absorption (Spanos *et al.*, 1978).



Growth hormone may also increase intestinal magnesium absorption (Hanna *et al.*, 1961) and cause a positive magnesium balance (Henneman *et al.*, 1960; Hanna *et al.*, 1961; Forbes *et al.*, 1962).

Phosphate balance is usually positive under the influence of growth hormone (Scriver and Bryan, 1935; Shorr *et al.*, 1953; Beck *et al.*, 1957; Ikkos *et al.*, 1958; Hutchings *et al.*, 1959; MRC, 1959; Beck *et al.*, 1960; Henneman *et al.*, 1960; Forbes *et al.*, 1962; Braithwaite, 1975). Although fecal phosphate has been found to be decreased in some studies (Beck *et al.*, 1957; Henneman *et al.*, 1960; Forbes *et al.*, 1962; Braithwaite, 1975), the major factor in the positive phosphate balance is the increased renal reabsorption of phosphate as discussed above.

### E. Interaction with Parathyroid Hormone

The serum calcium concentration is usually within the normal range in patients with acromegaly, although frank elevations have been reported (Molinatti *et al.*, 1961; Summers *et al.*, 1966; Nadarajah *et al.*, 1968; Brown and Singer, 1969; Aloia *et al.*, 1975). The cause of hypercalcemia in many patients may be related to an inherited predisposition for the development of multiple endocrine neoplasias. Primary hyperparathyroidism and acromegaly may coexist in this syndrome (Wermer, 1954). In two investigations it was noted that following successful treatment of the pituitary disease an elevated serum calcium concentration fell to normal in 12 patients (Nadarajah *et al.*, 1968; Brown and Singer, 1969). This is consistent with the hypothesis that growth hormone has a direct stimulatory effect on the parathyroid glands (Harrison *et al.*, 1960; Summers *et al.*, 1966). Hypophysectomy in the rat has been observed to result in a reduction in parathyroid gland size (Salzer, 1971; Fontaine *et al.*, 1978) and a decrease in serum immunoreactive parathyroid hormone concentration (Spanos *et al.*, 1978), although the serum calcium concentrations remained normal. The administration of bovine growth hormone to rats increased the serum calcium and immunoreactive parathyroid hormone concentrations as well as the weight of the parathyroid glands in one study (Lancer *et al.*, 1976). However, in another study human growth hormone raised the serum calcium concentration in rats, but with morphological evidence of decreased parathyroid gland function (Altenahr and Kampf, 1976). The morphologic changes were thought to be secondary to the hypercalcemia. The hyperphosphatemic and hypercalciuric effects of growth hormone may secondarily enhance parathyroid hormone secretion, although it has been observed that the hypercalciuric effect of growth hormone in the rat is prevented by parathyroidectomy (Fraser and Harrison, 1960). Measurement of the serum immunoreactive parathyroid hormone concentration in acromegaly has revealed an elevation in only one of six patients studied (Aloia *et al.*, 1975) and no change was seen following chronic growth hormone therapy to children

(Gertner *et al.*, 1979; Chipman *et al.*, 1980). Other indirect evidence of an interrelationship of growth hormone with parathyroid hormone is the observation that human growth hormone therapy shortens the recovery time of the blood calcium concentration following EDTA-induced hypocalcemia (Gershberg *et al.*, 1967).

#### F. Interaction with Calcitonin

The effect of growth hormone on calcitonin secretion and action has not been fully evaluated. Hypophysectomy in the rat resulted in a reduction in the size of parafollicular cell nuclei, cytoplasm, and granules (Salzer, 1971). In these rats calcium infusion resulted in prolonged hypercalcemia, a finding suggestive of calcitonin deficiency.

#### G. Interaction with Vitamin D

Recent studies have demonstrated an effect of growth hormone on vitamin D metabolism. Spencer and Tobiassen (1977) found that hypophysectomy decreased the amount of labeled  $1,25(\text{OH})_2\text{D}_3$  in the serum and intestinal mucosa in the rat following administration of labeled  $25(\text{OH})\text{D}_3$ . An increase in the levels of  $24,25(\text{OH})_2\text{D}_3$  was seen, reciprocal to the decrease in  $1,25(\text{OH})_2\text{D}_3$ . Fontaine *et al.* (1978) have also observed an increase in the conversion of  $25(\text{OH})\text{D}_3$  to  $24,25(\text{OH})_2\text{D}_3$  in the hypophysectomized rat. Growth hormone administration to the hypophysectomized rat partially reversed these changes. In other studies, hypophysectomized rats were shown to have a fall in plasma  $1,25(\text{OH})_2\text{D}_3$  concentration that returned to normal with the administration of human growth hormone (Spanos *et al.*, 1978; Pavlovitch *et al.*, 1978). Patients with acromegaly were found to have increased levels of plasma  $1,25(\text{OH})_2\text{D}$  and  $24,25(\text{OH})_2\text{D}$  which returned toward normal with bromocriptine therapy (Eskildsen *et al.*, 1979). The effect of growth hormone on vitamin D metabolism may not be direct, however, as growth hormone did not stimulate the formation of  $1,25(\text{OH})_2\text{D}_3$  *in vitro* (Bilke *et al.*, 1980).

The effect of growth hormone on intestinal calcium absorption may not be mediated only by an effect on vitamin D metabolism. The administration of growth hormone to growth hormone deficient children in two studies have shown that while an increase in growth, renal phosphate threshold, and fractional intestinal calcium absorption was seen, there was no change in plasma immunoreactive parathyroid hormone,  $25(\text{OH})\text{D}$ ,  $24,25(\text{OH})_2\text{D}$ , or urinary cyclic AMP (Gertner *et al.*, 1979; Chipman *et al.*, 1980). The level of  $1,25(\text{OH})_2\text{D}$  remained unchanged in one study (Gertner *et al.*, 1979) while falling in the other (Chipman *et al.*, 1980). In addition, growth hormone was demonstrated to increase intestinal calcium transport in the rat by a mechanism other than its influence on vitamin D metabolism (Aloia and Yeh, 1980).

## H. Summary

Growth hormone has a major influence on mineral metabolism, effected primarily through the intermediary action of the somatomedins. Growth hormone deficiency leads to dwarfism in the growing child, and an excess results in acromegaly and gigantism in the adult and child, respectively. Growth hormone causes an increase in both bone formation and resorption. While resorption may predominate in trabecular bone in man, bone accretion appears to predominate in the periosteal region. Bone mass during states of growth hormone excess is usually either normal or increased. A tendency toward hypercalcemia and hypercalciuria may result from increased bone resorption or through an increased intestinal calcium absorption. The enhanced absorption of dietary calcium may result from an effect of growth hormone on intestinal calcium transport or through an effect on vitamin D metabolism.

In addition to the above, growth hormone also raises the serum phosphate concentration. This is primarily due to an increase in the renal tubular phosphate threshold.

## V. PROLACTIN

Prolactin, a peptide hormone of the anterior pituitary gland, is important in the development of the mammary gland and in milk formation. It occurs in males, but its function is unknown. Large amounts of exogenous prolactin can simulate certain effects of growth hormone, including nitrogen retention. Some studies have suggested that prolactin plays a role in sodium and water metabolism, but this has not been a consistent finding. Studies of the mode of action of prolactin indicate that hormone binding to receptor sites on plasma membranes of mammary cells does not activate adenylate cyclase (Turkington *et al.*, 1973). Despite this, there is an increase of protein kinase activity and phosphorylated proteins in the cells.

A role of prolactin in mineral metabolism has been suggested by recent animal studies. During lactation there is an increased requirement for calcium that appears to be met by an increase in intestinal calcium absorption (Toverud *et al.*, 1976). It now appears that this adaptation is mediated by an increase in circulating levels of  $1,25(\text{OH})_2\text{D}_3$ , since levels of this vitamin D metabolite are up to fourfold greater in lactating than in normal rats (Boass *et al.*, 1977) and are twice normal in postpartum women (Kumar *et al.*, 1979). The rise in the concentration of  $1,25(\text{OH})_2\text{D}_3$  may be related to prolactin secretion, at least in part, since exogenous prolactin administration increases chick renal 25-hydroxycholecalciferol- $1\alpha$ -hydroxylase activity (Spanos *et al.*, 1976) and circulating levels of the metabolite in the chicken (Spanos *et al.*, 1976; Bikle *et al.*, 1980). Ample evidence has been obtained that exogenous prolactin increases active transport of calcium in the rat intestine (Finkelstein and Schachter, 1962;

Mainoya, 1975a,b). It is also possible that the rise in blood prolactin concentration found during pregnancy may contribute to the increased intestinal calcium absorption found in pregnant women (Heany and Skillman, 1971). This hypothesis is supported by the report of increased plasma  $1,25(\text{OH})_2\text{D}_3$  levels in pregnant women (Kumar *et al.*, 1979). However, other recent studies indicate that an effect of prolactin on vitamin D metabolism may be species-specific. Patients with high blood levels of prolactin due to pituitary tumors do not have elevation of circulating  $1,25(\text{OH})_2\text{D}$  (Adams *et al.*, 1979; Kumar *et al.*, 1979). Also prolactin failed to stimulate vitamin D metabolism in a study done in vitamin D-deficient rats (Matsumoto *et al.*, 1979).

Exogenous prolactin ( $10 \mu\text{g/hr}$ ) infused into normal, parathyroidectomized, or thyroparathyroidectomized rats raises plasma calcium concentration even in starved animals. In contrast, higher doses of prolactin ( $50 \mu\text{g/hr}$ ) reduce plasma calcium in intact rats (Robinson *et al.*, 1975). Injection of fish pituitary extracts into hypophysectomized hypocalcemic killifish produces a hypercalcemic effect also felt to represent a prolactin response (Pang *et al.*, 1978).

Injected prolactin has been shown to increase urinary calcium excretion in some (McAlister and Welbourn, 1962; Beck *et al.*, 1964; Robinson *et al.*, 1975) but not all subjects (Lipsett and Bergenstal, 1960; McAlister and Welbourn, 1962). Urinary calcium excretion in patients with endogenous hyperprolactinemia was not increased in one study (Hipkin *et al.*, 1978).

It would appear that prolactin also interacts with mineral metabolism as a consequence of an effect on serum estradiol levels. In a study of young amenorrheic hyperprolactinemic women serum estradiol levels were either at postmenopausal levels or at the levels found during the follicular phase of the menstrual cycle (Klibanski *et al.*, 1980). Bone density was reduced in these patients as compared to age-matched normal subjects and there was a positive correlation between serum estradiol concentrations and bone density.

## VI. INSULIN

### A. Mode of Action

Insulin has profound effects not only on carbohydrate metabolism but also on fat and protein metabolism. The control of blood sugar is modulated by the action of insulin to increase glucose transport into liver (by stimulating glucokinase), adipose cells, and muscle, by stimulation of glycogen synthesis in liver and muscle and by suppressing liver gluconeogenesis. Insulin influences fat metabolism by increasing lipogenesis in the liver and by decreasing lipolysis and stimulating fatty acid synthesis in adipose cells. The effect of insulin on protein metabolism is critical in that it stimulates increased amino acid uptake and

protein synthesis in muscle and inhibits proteolysis in the liver. It is an anabolic hormone and is essential for normal growth. Insulin does not appear to activate membrane-bound adenylate cyclase as an initiating step in its biological actions. It is likely that the action of insulin is initiated by binding to receptors on cell membranes, but the details of the subsequent messengers that are activated by the hormone have yet to be elucidated (Czech, 1977).

## B. Interaction with Bone

*In vitro* studies indicate that insulin can influence bone metabolism directly. Insulin enhances the incorporation of pyrimidine nucleosides into intracellular nucleotide and RNA pools in isolated bone cells (Peck and Messinger, 1970). An increase in membrane transport of amino acids in fetal rat calvaria has also been observed (Hahn *et al.*, 1971). The effect on amino acid membrane transport can be blocked by either puromycin or cycloheximide, suggesting that this phenomenon is mediated by synthesis of a membrane carrier protein. Bone collagen synthesis was increased by insulin in cultures of fetal rat or chick bone (Perlish *et al.*, 1973; Canalis *et al.*, 1977a) and newborn rat bone (Wettenhall *et al.*, 1969; Schwartz *et al.*, 1970). This could be demonstrated at concentrations as low as  $10^{-9}M$ , a level only twice that of the normal rat serum concentration (Canalis *et al.*, 1977a). The increase in collagen synthesis does not appear to reflect a generalized stimulation of cell growth, since collagen synthesis was found to be greater than noncollagen protein synthesis.

*In vivo* studies in rats have also provided strong evidence for an effect of insulin on bone metabolism. Diabetic rats have reduced incorporation of calcium and citrate into bone which is corrected by insulin administration (Dixit and Stern, 1979). A reduction in osteoid, a failure to acquire a tetracycline label, reduced numbers of osteoclasts and a reduction of bone alkaline phosphatase has been demonstrated in a second study of diabetic rats (Shires *et al.*, 1981). Insulin has also been shown to stimulate matrix-induced cartilage and bone formation (Weiss and Reddi, 1980).

Evidence to support abnormal metabolic activity of bone in diabetic patients has been minimal. An increase in total serum alkaline phosphatase activity is a feature often found in diabetes mellitus (Heath *et al.*, 1979). This finding has usually been attributed to release of liver or intestinal alkaline phosphatase. In the only study of the activity of the bone isoenzyme, increased activity was found in patients on dietary, drug, and insulin regimens (Štěpán *et al.*, 1980).

The above studies should lead one to expect that insulin deficiency in humans would result in decreased bone mass. This appears to be the case in children with diabetes mellitus. Three studies utilizing the Norland-Cameron Bone-Mineral Analyzer have shown a significant decrease in bone mass of the forearm as compared to sex- and age-matched controls (Levin *et al.*, 1976; Rosenbloom

*et al.*, 1977; McNair *et al.*, 1978, 1979). In one study, the greatest abnormality was found in white females (Rosenbloom *et al.*, 1977). Levin *et al.* (1976) commented that the abnormality probably reflects the underlying disease and not the effects of insulin deficiency, since they found no correlation of severity with insulin requirements or duration of the disease. The opposite conclusion was reached by Rosenbloom *et al.* (1977) who found a much lower frequency of decreased bone mass in white females after 5 years of insulin therapy. McNair *et al.* (1978, 1979), in the largest study, found that the loss of bone mass appeared to coincide with the onset of clinical diabetes mellitus and that the deficit was greater in patients whose disease developed before age 20 than after age 25. They also found that the bone mineral content was inversely correlated with the fasting blood glucose, glycosuria, and the insulin requirement. Elevated urinary excretion of calcium was common during the early course of the disease and correlated with the loss of skeletal mineral. No histologic studies of bone from diabetic children have been reported.

Studies in adults with diabetes mellitus have produced quite variable results. Although some patients may have a decreased bone mass (Hernberg, 1952; Levin *et al.*, 1976; Ringe *et al.*, 1976; DeLeeuw and Abs, 1977; McNair *et al.*, 1978), many patients have no demonstrable loss of bone by radiologic (Meema and Reid, 1969; Forgács *et al.*, 1976),  $^{125}\text{I}$  photon-absorption (DeLeeuw and Abs, 1977), or histologic evaluation (Kelin and Frost, 1964; DeLeeuw *et al.*, 1976). In fact, it has been shown that an increased bone mass is not an uncommon finding (Kelin and Frost, 1964; Meema and Reid, 1969; DeLeeuw *et al.*, 1976; Forgács *et al.*, 1976; DeLeeuw and Abs, 1977). It is likely that several factors account for the variability in findings of the above studies. The age of onset of diabetes mellitus, the degree of insulin deficiency, the duration of therapy, the type of therapy (insulin versus sulfonylurea compounds), the methods used to evaluate bone mass, and other environmental and genetic influences undoubtedly interact to produce a wide range of bone architecture in the diabetic patient. Despite the recent interest in the bone density of diabetic patients, it is still unclear as to whether these patients are susceptible to fracture. A large survey in Rochester, Minnesota revealed that the risk of skeletal fracture in diabetics was not increased (Heath *et al.*, 1980).

### C. Interaction with Kidney

In experimental diabetes (Weber *et al.*, 1976) and in patients with diabetic ketoacidosis (Seldin and Tarail, 1950; Raskin *et al.*, 1978; McNair *et al.*, 1979), hypercalciuria and/or hyperphosphaturia have been noted. It has been suggested that associated metabolic acidosis and osmotic diuresis are responsible for these findings. It is also possible that relative insulin deficiency could account for hyperphosphaturia since phosphate excretion is decreased by insulin infusion in

normal subjects (DeFronzo *et al.*, 1975). In the same study urinary calcium excretion was increased during insulin infusion.

#### D. Interaction with Intestine

A large body of evidence has accumulated that points to a role of insulin in the intestinal transport of calcium in experimental animals. Either alloxan or streptozotocin-induced diabetes in the rat produces reduced duodenal calcium absorption (Schneider and Schedl, 1972, 1977; Schneider *et al.*, 1975, 1976, 1977a), which can be restored by insulin treatment (Schneider *et al.*, 1977a). An associated abnormality in experimental diabetes is a decrease in duodenal calcium-binding protein (Schneider and Schedl, 1977). It has been demonstrated that either  $1,25(\text{OH})_2\text{D}_3$  or  $1\alpha$ -hydroxycholecalciferol is capable of returning duodenal calcium transport to normal (Schneider *et al.*, 1976). In contrast, vitamin  $\text{D}_3$  or  $25(\text{OH})\text{D}_3$  administration failed to alter calcium transport in this study. These observations provide indirect evidence that decreased calcium transport arises as a result of an abnormality in  $1\alpha$ -hydroxylation of  $25(\text{OH})\text{D}_3$ . Direct evidence of the role of vitamin D in the calcium malabsorption associated with experimental diabetes mellitus was provided by the observation that diabetic rats have serum  $1,25(\text{OH})_2\text{D}_3$  concentrations one-eighth to one-half that of normal rats (Schneider *et al.*, 1977b; Shires *et al.*, 1981). Insulin treatment restores the concentration to normal. These results do not distinguish between impaired synthesis or increased catabolism as the pathogenesis of the low serum  $1,25(\text{OH})_2\text{D}_3$  concentration. A study of the synthesis and metabolic clearance of this metabolite in diabetic rats suggested that the action of insulin was to stimulate renal  $25$ -hydroxyvitamin  $\text{D}_3$ - $1\alpha$ -hydroxylase activity (Spencer *et al.*, 1980). Insulin permitted parathyroid hormone stimulation of  $1,25(\text{OH})_2\text{D}_3$  production in cultured chick kidney cells but did not have an effect alone (Henry, 1981).

In three studies of experimental diabetes in the rat, parathyroid hormone concentration in the circulation was found to be increased (Schneider *et al.*, 1974; Weber *et al.*, 1976; Schedl *et al.*, 1978). This evidence of secondary hyperparathyroidism is consistent with a homeostatic attempt to stimulate renal  $25$ -hydroxyvitamin  $\text{D}_3$   $1\alpha$ -hydroxylase activity and restore intestinal calcium absorption to normal. Inexplicably, low levels of parathyroid hormone were reported in one study of diabetic rats (Shires *et al.*, 1981).

In one study of intestinal magnesium absorption in diabetic rats there was a decrease in duodenal magnesium absorption when expressed per gram dry weight of mucosa (Miller and Schedl, 1976). However, the absorptive capacity of a given segment-length of duodenum was unchanged due to a marked increase of mucosal growth in diabetic animals. This finding contrasts with the studies of calcium absorption in that both specific absorption and segment absorptive capacity of calcium are both reduced. The difference between magnesium and

calcium absorption in diabetic rats suggests an inherent difference in the regulation of transport of these divalent cations in the rat.

The relevance of the animal studies to the osteopenia of human diabetes has been challenged by a study comparing 40 healthy subjects with 82 diabetic patients (Heath *et al.*, 1979). No differences in intestinal calcium absorption, serum  $1,25(\text{OH})_2\text{D}$  concentration or serum parathyroid hormone concentration were detected.

## E. Summary

A number of *in vitro* studies have demonstrated the anabolic effects of insulin on bone. Insulin enhances incorporation of pyrimidine nucleosides into bone cell nucleotide and RNA pools, stimulates membrane transport of amino acids in fetal rat calvaria and stimulates bone collagen synthesis. In children with diabetes mellitus a decrease in bone mass has been observed. This finding is less predictable in adults. Studies in experimental animals have focused on the observation that drug-induced diabetes mellitus produces reduced intestinal calcium absorption that is reversed by insulin treatment. Serum  $1,25(\text{OH})_2\text{D}_3$  levels are reduced in the diabetic animals and return to normal after insulin treatment.

## VII. GLUCAGON

### A. Mode of Action

Glucagon is a peptide hormone whose main physiologic role is the control of blood glucose concentration. The hyperglycemic effect of this hormone is mediated by stimulation of hepatic glycogenolysis and gluconeogenesis. These effects are initiated by binding of glucagon to receptors on hepatic cell plasma membranes and activation of membrane-bound adenylate cyclase (Pohl, 1977).

Evidence that glucagon may be a factor in mineral metabolism is provided by the observation that glucagon administration produces transient hypocalcemia in the dog, rabbit, rat, sheep, and man (Paloyan *et al.*, 1967a,b; Aliapoulos and Morain, 1968; Bowser *et al.*, 1968; Avioli *et al.*, 1969; Barlet, 1969; Birge and Avioli, 1969; Ottenjann and Deyhle, 1969; Pickleman *et al.*, 1969; Tanzer *et al.*, 1970; Williams *et al.*, 1969; Federspil *et al.*, 1970; Hattner *et al.*, 1970; Raman, 1970b; Ulbrych-Jablonska, 1972; Aliapoulos *et al.*, 1973). It has also been suggested that endogenous glucagon release in patients with acute pancreatitis may be primarily responsible for the hypocalcemia encountered in such patients (Paloyan *et al.*, 1976b). Data concerning the mechanism by which glucagon induces hypocalcemia will now be considered.



## B. Interaction with Bone

The hypocalcemic effect of glucagon appears to arise primarily from a direct action on the skeleton, although there is also evidence of an indirect action. The known stimulation of calcitonin secretion by glucagon (see Chapter 8, this volume) suggested that the hypocalcemia was due to a decrease in bone resorption mediated by calcitonin. Evidence to support this hypothesis was provided by studies in dogs (Avioli *et al.*, 1969) and rabbits (Pickleman *et al.*, 1969) where thyroidectomy was found to prevent a hypocalcemic response to glucagon. In contrast, four studies in rats clearly indicated that the hypocalcemic response was not dependent on the presence of the thyroid gland (Williams *et al.*, 1969; Hattner *et al.*, 1970; Stern and Bell, 1970; Tanzer *et al.*, 1970). The hypocalcemic response to glucagon in humans is dependent to a great extent on the serum calcium concentration, a larger decrease being found in hypercalcemic patients and a smaller or no decrease being found in normocalcemic subjects (Paloyan *et al.*, 1967a; Birge and Avioli, 1969; Ottenjann and Deyhle, 1969; Federspil *et al.*, 1970; Ulbrych-Jablonska, 1972). Of interest also is the finding that glucagon infusions in normocalcemic or hypocalcemic patients with pancreatitis produce no significant fall in serum calcium concentration (Condon *et al.*, 1975; Robertson *et al.*, 1976). No studies in athyroidal patients have been reported.

A direct effect of glucagon on bone *in vitro* has been reported in two studies. Stern and Bell (1970) found that glucagon inhibited bone resorption in embryonic rat bone cultures stimulated with either parathyroid hormone or dibutyryl-3',5'-adenosine monophosphate. No effect could be found in unstimulated cultures, an observation opposite to that found with calcitonin (see Chapter 8, this volume). Gozariu *et al.* (1971) reported identical results in a study utilizing newborn rat bone in culture. Kalu *et al.* (1972) studied the effect of administered glucagon on bone collagen metabolism in the thyroparathyroidectomized rat and also concluded that glucagon inhibited parathyroid hormone-induced bone resorption but had little effect in normal rats.

Williams *et al.* (1969) compared the effects of calcitonin and glucagon on  $^{47}\text{Ca}$  kinetics and suggested that glucagon produced hypocalcemia by promoting uptake of  $^{47}\text{Ca}$  by bone. A subsequent study demonstrated that glucagon administration produced increased femoral weight in control and immobilized rat extremities, although a relative degree of osteoporosis was present in the immobilized extremity (Woodward and Jowsey, 1972). This observation would tend to support the concept of glucagon as an inducer of bone formation. However, Kalu *et al.* (1972) found that glucagon inhibited incorporation of proline into rat bone *in vivo* and Canalis *et al.* (1977a) reported that glucagon inhibited proline incorporation into embryonic rat bone *in vitro*. In man, administration of glucagon may decrease bone resorption and bone formation as evidenced by a rapid decrease in urinary hydroxyproline excretion and plasma alkaline phos-

phatase concentration in patients with Paget's disease treated with glucagon (Condon, 1971). These effects were not found in a subsequent study, possibly because the glucagon doses administered were somewhat lower (Singer *et al.*, 1978). In general, very large doses of glucagon are required to produce effects on mineral metabolism. This suggests that this hormone may not play an important physiologic role in the regulation of mineral metabolism.

### C. Interaction with Kidney

It has been found in the dog that pharmacologic doses of glucagon induce significant increases in urinary calcium excretion (Charbon *et al.*, 1963; De-wonck *et al.*, 1963; Pullman *et al.*, 1967). This has not been confirmed in man (Birge and Avioli, 1969). Since nephrectomy did not abolish the hypocalcemic effect of glucagon in any species studied (Avioli *et al.*, 1969; Williams *et al.*, 1969; Tanzer *et al.*, 1970; Aliapoulos *et al.*, 1973), it is unlikely that the hypocalcemic effect is dependent on renal calcium excretion.

Glucagon administration also produces hypophosphatemia of a significant degree (Avioli *et al.*, 1969; Tanzer *et al.*, 1970; Aliapoulos *et al.*, 1973). Increased renal excretion of inorganic phosphate has been found after glucagon administration in dogs and humans (Staub *et al.*, 1957; Elrick *et al.*, 1958; Charbon *et al.*, 1963; Pullman *et al.*, 1967). In two studies undertaken in nephrectomized rats, contradictory results were obtained concerning the role of the kidney in glucagon-induced hypophosphatemia. Tanzer *et al.* (1970) found that prior nephrectomy did not alter the hypophosphatemic response, whereas Aliapoulos *et al.* (1973) observed a diminution in hypophosphatemia following nephrectomy.

### D. Interaction with Intestine

Little is known of the effect of glucagon on intestinal transport of calcium. A study in rats demonstrated that removal of the entire gastrointestinal tract, but not the stomach alone, decreased the hypocalcemic effect of glucagon (Aliapoulos *et al.*, 1973). In this study, intestinal transport of calcium was not assessed, so that the possibility of a direct effect of glucagon on calcium transport was not substantiated. A study of the effect of glucagon on colonic radiocalcium excretion in rats revealed that fecal radiocalcium was reduced tenfold in glucagon-treated rats as compared to control rats given systemic  $^{45}\text{Ca}$  (Bogdat *et al.*, 1974). The latter study does not support the hypothesis that the intestine is involved in the hypocalcemic response to glucagon.

### E. Summary

Glucagon, in pharmacologic doses, is capable of producing transient hypocalcemia in a variety of species. This response is dependent on the presence of the

thyroid gland in dogs and rabbits but not in rats. In rats, removal of the entire gastrointestinal tract reduced the hypocalcemic response. *In vitro* studies have demonstrated a direct effect of glucagon on bone resorption. Parathyroid hormone-induced bone resorption was inhibited by glucagon in two studies, although no effect was found on unstimulated bone. The physiologic significance of these observations is unclear.

## VIII. GASTROINTESTINAL HORMONES

### A. Gastrin

Gastrin is a peptide hormone of considerable structural heterogeneity found in mucosal cells of the gastric antrum, in the D cells of the pancreas and in the mucosa of the small intestine, primarily the proximal duodenum. The hormone is a potent secretagogue of gastric acid and affects esophageal motility, as well as muscle contraction of the stomach, small and large intestine, gallbladder, and uterus. The mode of action of the hormone is unknown; adenylate cyclase activation is not the mediator of its biological effects.

The administration of gastrin to pigs, lambs, and rats produces a significant degree of hypocalcemia (Care *et al.*, 1971; Cooper *et al.*, 1971, 1972; Barlet, 1973; Schulak and Kaplan, 1974, 1975; Kaplan *et al.*, 1977; Krishnamra and Limlomwongse, 1978). Since it was known that gastrin or pentagastrin could stimulate calcitonin release (see Chapter 8, this volume), it was initially thought that the hypocalcemic effect of gastrin was mediated by calcitonin. This speculation was supported by the observations that thyroidectomy abolished the hypocalcemic effect in the pig (Care *et al.*, 1971), in the lamb (Barlet, 1973), and in the rat (Cooper *et al.*, 1972). In subsequent studies in which younger rats were used and/or higher doses of gastrin were administered, the hypocalcemic effect persisted in the absence of the thyroid (Schulak and Kaplan, 1974, 1975; Kaplan *et al.*, 1977; Krishnamra and Limlomwongse, 1978). In the studies of Schulak and Kaplan, prior total gastrectomy or removal of the proximal 75% of the stomach abolished the hypocalcemic effect of gastrin. An increase in gastric calcium excretion has been observed in response to gastrin administration and could, in part, account for the role of the stomach (Krishnamra and Limlomwongse, 1978). Increased fecal calcium excretion after gastrin administration has been found in normal and thyroidectomized sheep (Barlet, 1973). It is not known what was the source of the fecal calcium in this study. Synthetic tetragastrin increased calcium absorption *in vivo* in a rat duodenal loop study (Fujita, 1973). This was found in intact but not parathyroidectomized animals.

Stimulation of endogenous hypergastrinemia by gastric antral exclusion in rats does not result in hypocalcemia (Grimelius *et al.*, 1977). This could be due to the induction of parathyroid hyperplasia observed in the animals. Parathyroid hor-

mone assays were not done in this study. In sheep 1-hr infusions of gastrin failed to stimulate parathyroid hormone secretion (Jaffe *et al.*, 1974).

Gastrin does not increase urinary calcium excretion (Barlet, 1973; Krishnamra and Limlomwongse, 1978) and nephrectomy does not abolish its hypocalcemic effect in rats (Schulak and Kaplan, 1975).

Cholecystokinin-pancreozymin, which contains the same carboxyl-terminal four amino acids as gastrin, is also a hypocalcemic agent in lambs (Barlet and Garel, 1976).

## B. Secretin

Secretin, the first hormone to be recognized, is a peptide found in the proximal small intestine. Administration of this hormone results in increased bicarbonate and water secretion by the exocrine pancreas. Secretin also acts on the endocrine pancreas through a stimulatory effect on insulin secretion. The action of the hormone appears to be mediated through activation of adenylate cyclase (Schwartz *et al.*, 1974).

There is no evidence of an effect of endogenous secretin on calcium metabolism, but observation of an increase in serum calcium concentration in a patient with Zollinger-Ellison syndrome given secretin prompted further studies (Isenberg *et al.*, 1972). Infusion of secretin in normal subjects and patients with Zollinger-Ellison syndrome and primary hyperparathyroidism produced a significant increase in total and ionized serum calcium in one study (Isenberg *et al.*, 1973). A subsequent study in normal subjects, duodenal ulcer patients, and patients with primary hyperparathyroidism revealed a rise in total calcium but not ionized calcium concentration during secretin infusion (Schwille *et al.*, 1975). Two other studies in man (J. C. Thompson *et al.*, 1972; Bradley *et al.*, 1975) and one in the rat (Kaplan *et al.*, 1977) failed to demonstrate any significant hypercalcemia due to secretin. It is unclear as to why there is such variability in the results. An elegant *in vitro* study of the effect of secretin on isolated bovine parathyroid cells did demonstrate a simultaneous rise in intracellular cyclic AMP accumulation and parathyroid hormone release (Windeck *et al.*, 1978).

## C. Vasoactive Intestinal Polypeptide

Vasoactive intestinal polypeptide (VIP) is a 28-amino-acid putative hormone found in the intestine and nervous system that has been demonstrated to have pharmacologic effects on the cardiovascular, respiratory, and gastrointestinal systems. The gastrointestinal effects of this peptide appear to be mediated by cyclic AMP and include increased fecal loss of water and electrolytes and achlorhydria (Schwartz *et al.*, 1974). These findings are common in a syndrome termed Verner-Morrison Syndrome (Verner and Morrison, 1958), pancreatic

cholera, watery diarrhea syndrome, or the syndrome of watery diarrhea, hypokalemia, and achlorhydria. Most of the patients with this syndrome have a pancreatic islet cell tumor. Recent studies have indicated that VIP may be a factor in the pathogenesis of this syndrome, since elevated serum and tissue concentrations of VIP have been found in most patients (Bloom *et al.*, 1973; Rambaud *et al.*, 1975; Gagel *et al.*, 1976; Shields and Haff, 1976; Holdaway *et al.*, 1977; Bloom, 1978; Cooperman *et al.*, 1978). Other diarrheogenic agents such as prostaglandin E have also been implicated in the pathogenesis of this syndrome (Jaffe *et al.*, 1977).

Hypercalcemia has been observed in some patients with the Verner-Morrison syndrome and is reversible when the pancreatic tumor is resected (Verner and Morrison, 1958, 1974; Holdaway *et al.*, 1977). It is not certain whether VIP is responsible for the abnormal calcium homeostasis. In a preliminary report, VIP injection raised serum calcium concentrations by 5% after 10 min in dogs (Makhlouf *et al.*, 1974). However, in the rat VIP caused no detectable hypercalcemia (Kaplan *et al.*, 1977). It has also been reported that prostaglandin E (Jaffe *et al.*, 1977) and parathyroid hormone (Hirose *et al.*, 1975) may be involved in the pathogenesis of the hypercalcemia. Since there also are patients with hypercalcemia and low serum PTH concentrations (Rambaud *et al.*, 1975; Gagel *et al.*, 1976; Holdaway *et al.*, 1977) and patients with elevated plasma prostaglandin E levels who are normocalcemic (Rambaud *et al.*, 1975), further studies will be required to determine the pathogenesis of hypercalcemia in this syndrome.

#### D. Summary

The gastrointestinal hormones gastrin, secretin, and vasoactive intestinal peptide have all been reported to exert an influence on extracellular fluid calcium concentration. Gastrin administration to a variety of species produces transient hypocalcemia. Secretin has been reported to raise the serum calcium concentration in two studies, but other studies have failed to confirm the hypercalcemic effect. Hypercalcemia is sometimes found in patients with the syndrome of watery diarrhea, hypokalemia, and achlorhydria. These patients usually harbor a pancreatic tumor and have elevated serum and tumor vasoactive intestinal peptide concentrations. The physiologic significance of the above observations is unknown.

### IX. EPINEPHRINE

Epinephrine, the catecholamine primarily synthesized in the adrenal medulla, interacts with beta adrenergic receptors in numerous organs and thereby influences the regulation of blood flow, blood glucose and free fatty acid concen-

trations, renal sodium reabsorption, bowel, and bladder tone and body temperature. These actions of epinephrine are associated with activation of adenylate cyclase.

Interest in the role of epinephrine in mineral metabolism has existed for more than 50 years since it was first recognized that epinephrine could influence calcium concentration in the circulation (Kenny, 1964). As Kenny has pointed out, the early studies produced conflicting results, with both hypercalcemia and hypocalcemia reported to have resulted from epinephrine administration. Studies during the past 15 years have to some extent helped clarify the complex interaction between epinephrine and extracellular calcium regulation.

Administration of epinephrine or other  $\beta$ -adrenergic agonists causes an acute increase in parathyroid hormone (see Chapter 6, this volume) and calcitonin (see Chapter 8) secretion in several species including man. In intact animals there is usually no detectable change in blood calcium concentration after  $\beta$ -adrenergic stimulation (Kenny, 1964; Morey and Kenny, 1964; Wells and Lloyd, 1968; Fischer *et al.*, 1973; Hsu and Cooper, 1975; Kukreja *et al.*, 1975; Blum *et al.*, 1978; Harney *et al.*, 1978; Metz *et al.*, 1978), although a small degree of hypocalcemia has sometimes been reported (Phillippo *et al.*, 1970; Fischer *et al.*, 1973; Vora *et al.*, 1978). A simultaneous rise in calcitonin secretion may prevent a hypercalcemic response (Harney *et al.*, 1978). In parathyroidectomized animals  $\beta$ -adrenergic agonists can elevate blood calcium concentration, whether or not the thyroid gland is present (Kenny, 1964; Morey and Kenny, 1964; Wells and Lloyd, 1968; Hsu and Cooper, 1975). This suggests that calcitonin secretion is not solely responsible for blocking a parathyroid-induced hypercalcemic response to epinephrine in intact animals. It may be that a prevailing normal rate of bone resorption in intact animals does not permit the hypercalcemic action of  $\beta$ -adrenergic stimulation to be expressed.

There are 15 case reports of patients with pheochromocytoma and hypercalcemia in whom adrenalectomy was reported to reverse the hypercalcemic state (Swinton *et al.*, 1972; Kukreja *et al.*, 1973; Bouillon *et al.*, 1974; Finlayson and Casey, 1975; Ghose *et al.*, 1976; Gray and Gillon, 1976; Kalager *et al.*, 1977; Atuk *et al.*, 1979). Parathyroid hormone concentrations in the circulation were measured in eight patients and found to be elevated or inappropriately detectable in seven patients. In no patient was an arteriovenous gradient of parathyroid hormone concentration measured in an attempt to localize the site of abnormal parathyroid hormone secretion. Also no tumor was examined for parathyroid hormone content. In a group of ten normocalcemic patients with pheochromocytoma, serum parathyroid hormone concentrations were normal (Miller *et al.*, 1975). In man it seems unlikely that a chronic excess of catecholamines in the circulation leads to hypercalcemia. The hypercalcemia reversed by adrenalectomy in patients with pheochromocytoma appears to result from ectopic secretion of parathyroid hormone or other hypercalcemic agents.

The hypercalcemic effect of epinephrine in parathyroidectomized animals probably can be explained by a direct effect of the hormone on bone. A renal mechanism is ruled out by the observation that epinephrine increases urinary calcium in parathyroidectomized rats (Morey and Kenny, 1964). A gastrointestinal effect also is unlikely since rats that have fasted for 18–24 hr still manifest hypercalcemia (Hsu and Cooper, 1975). Consistent with the hypothesis of a direct skeletal action of epinephrine are the observations that cyclic 3'5' adenosine monophosphate content is increased by epinephrine in isolated bone cells (Peck *et al.*, 1974; Rodan and Rodan, 1974) and in fetal rat calvaria (Chase and Aurbach, 1970). It has yet to be reported that epinephrine increases bone resorption *in vitro*. This information is needed to help establish the mechanism by which  $\beta$ -adrenergic agonists can increase extracellular calcium concentration.

## X. ERYTHROPOIETIN

Erythropoietin is a glycoprotein with a molecular weight of approximately 60,000. It is thought to be synthesized by the kidney and/or by the liver in response to a renal factor and to be released into the circulation in response to blood loss or a lowering of plasma oxygen tension. The hormone acts to stimulate erythropoiesis and release red cells from the bone marrow into the systemic circulation (Gordon, 1973).

Perris and Whitfield (1971) observed that acute blood loss in rats resulted in significant hypercalcemia beginning 24 hr after induction of hemorrhage. The hypercalcemia persisted for several days and returned to normal coincident with correction of the anemia. Thyroparathyroidectomy prior to hemorrhage prevented the hypercalcemic response and also the rapid restoration of red cell mass. Since significant hypocalcemia occurred during the initial 24 hr after hemorrhage, it was hypothesized that an overshoot of parathyroid hormone secretion in response to the early hypocalcemia might explain the transient hypercalcemia. However, subsequent studies utilizing exogenous erythropoietin in rats suggest a role of this hormone in calcium homeostasis (Hunt and Perris, 1973). Exogenous hormone administered subcutaneously produced a rise of plasma calcium concentration of the order of 1 mg/dl. This hypercalcemic effect was abolished by prior parathyroidectomy. Since parathyroid hormone levels were not measured, it is not known whether erythropoietin is capable of stimulating parathyroid hormone secretion or whether it potentiates the peripheral actions of the hormone.

There is little evidence to support the role of erythropoietin in the physiology or pathophysiology of calcium homeostasis in man. In one unconfirmed study of children with congenital heart disease and elevated hematocrits, ionized calcium concentrations were elevated (Putnam, 1972).

## XI. SUMMARY

Many hormones whose principal actions are not directed towards mineral metabolism do contribute in a minor or even major fashion to mineral homeostasis. These hormones may act directly on the skeleton, intestine, and kidney or may influence the function of these organs by interacting with the major hormonal regulators of mineral metabolism.

Steroid hormones have a profound effect on the skeleton. An excess of endogenous cortisol or pharmacologic doses of glucocorticosteroids may produce a marked deficit in bone mass. Deficiency of estrogens or testosterone also contributes to the development of osteoporosis.

Thyroid hormone directly influences bone metabolic activity. In hyperthyroidism hypercalcemia and/or osteoporosis may occur when the rate of bone resorption exceeds that of bone formation. The main skeletal manifestation of hypothyroidism is failure of epiphyseal development leading to short stature.

Growth hormone, through the intermediary action of somatomedin, is the critical hormonal regulator of growth. Dwarfism and gigantism or acromegaly may result from deficiency or excess of the hormone, respectively. Growth hormone also induces an increase in intestinal calcium absorption and may produce hypercalcemia, possibly by a stimulation of  $1,25(\text{OH})_2\text{D}$  production.

Prolactin may be responsible for the increased intestinal calcium absorption found in lactating women and perhaps in pregnant women. Prolactin also is capable of stimulating  $1,25(\text{OH})_2\text{D}$  production in animals.

The pancreatic hormones insulin and glucagon affect mineral homeostasis in differing degrees. Insulin is required for normal bone development in children. In animals, insulin deficiency results in decreased intestinal calcium absorption in association with low levels of  $1,25(\text{OH})_2\text{D}_3$ . Glucagon, in pharmacologic doses, produces transient hypocalcemia, but its physiologic role is unknown.

Gastrin, secretin, and vasoactive intestinal peptide have been shown to alter the extracellular calcium concentration in animals but the physiologic role of these gastrointestinal hormones is not clear.

Epinephrine can produce hypercalcemia in parathyroidectomized animals, but this has not been demonstrated in man.

Erythropoietin has been reported to raise the serum calcium concentration in rats. Prior parathyroidectomy prevents the response. Definitive studies in man are lacking.

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